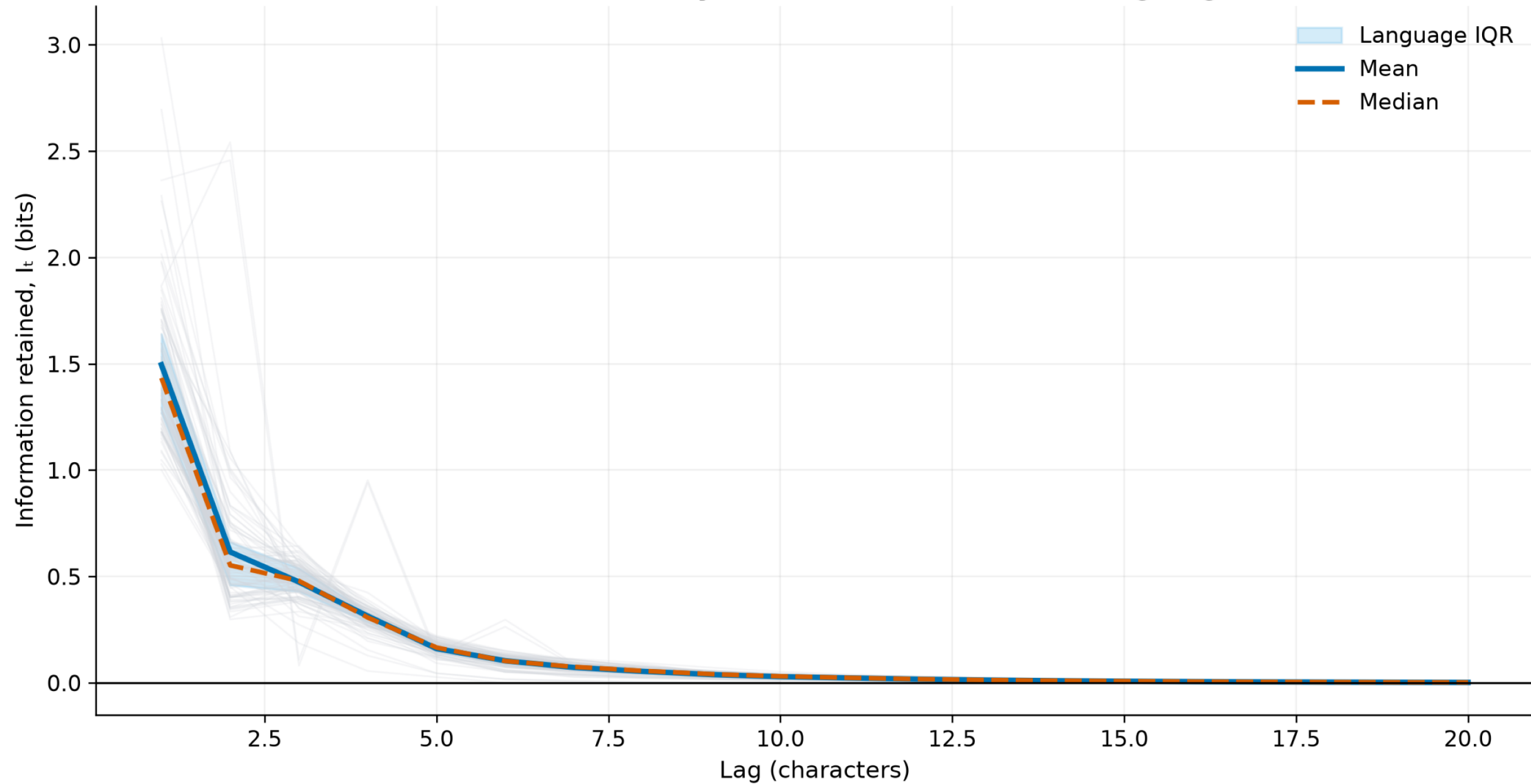
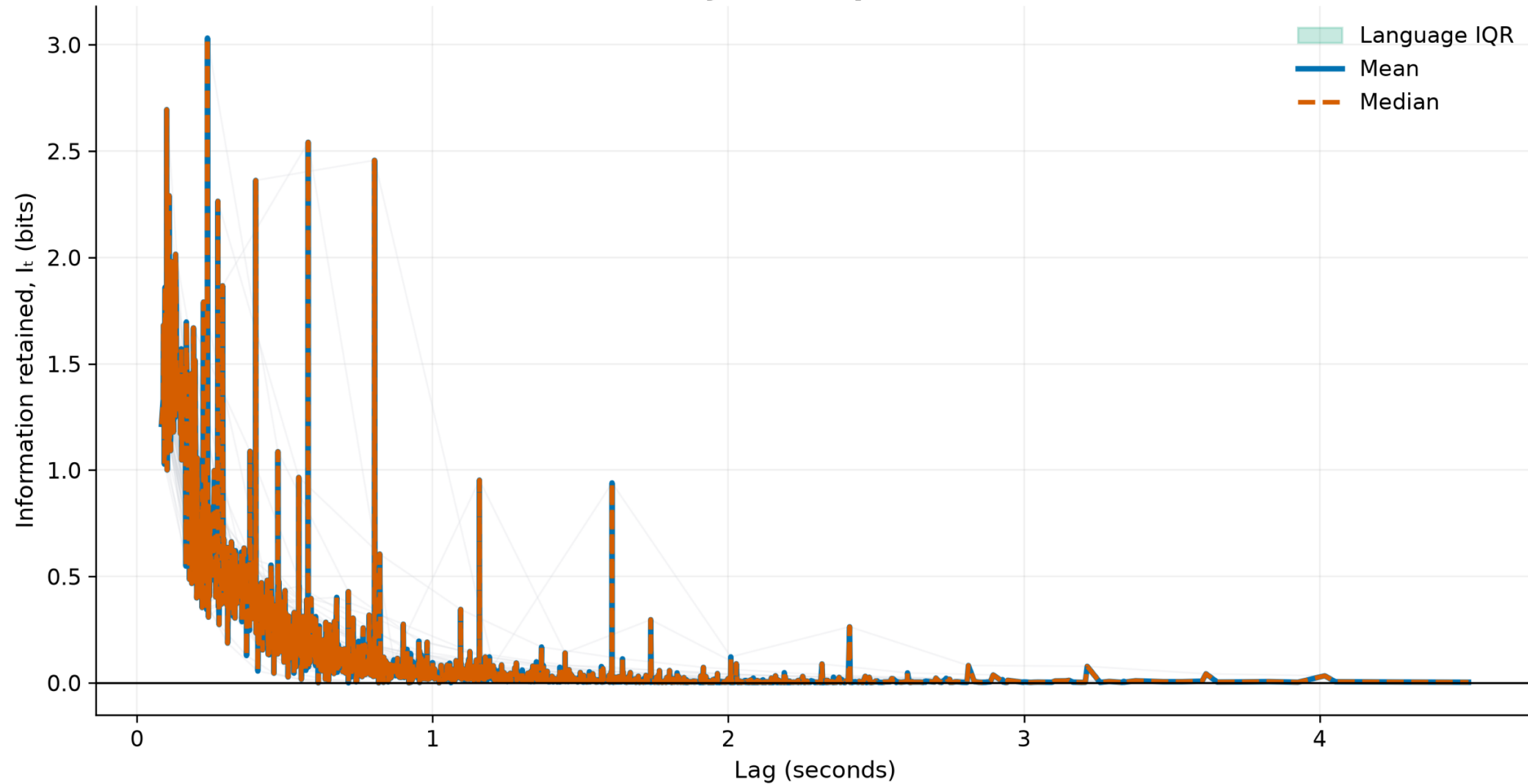


# Information locality across 102 FLEURS languages

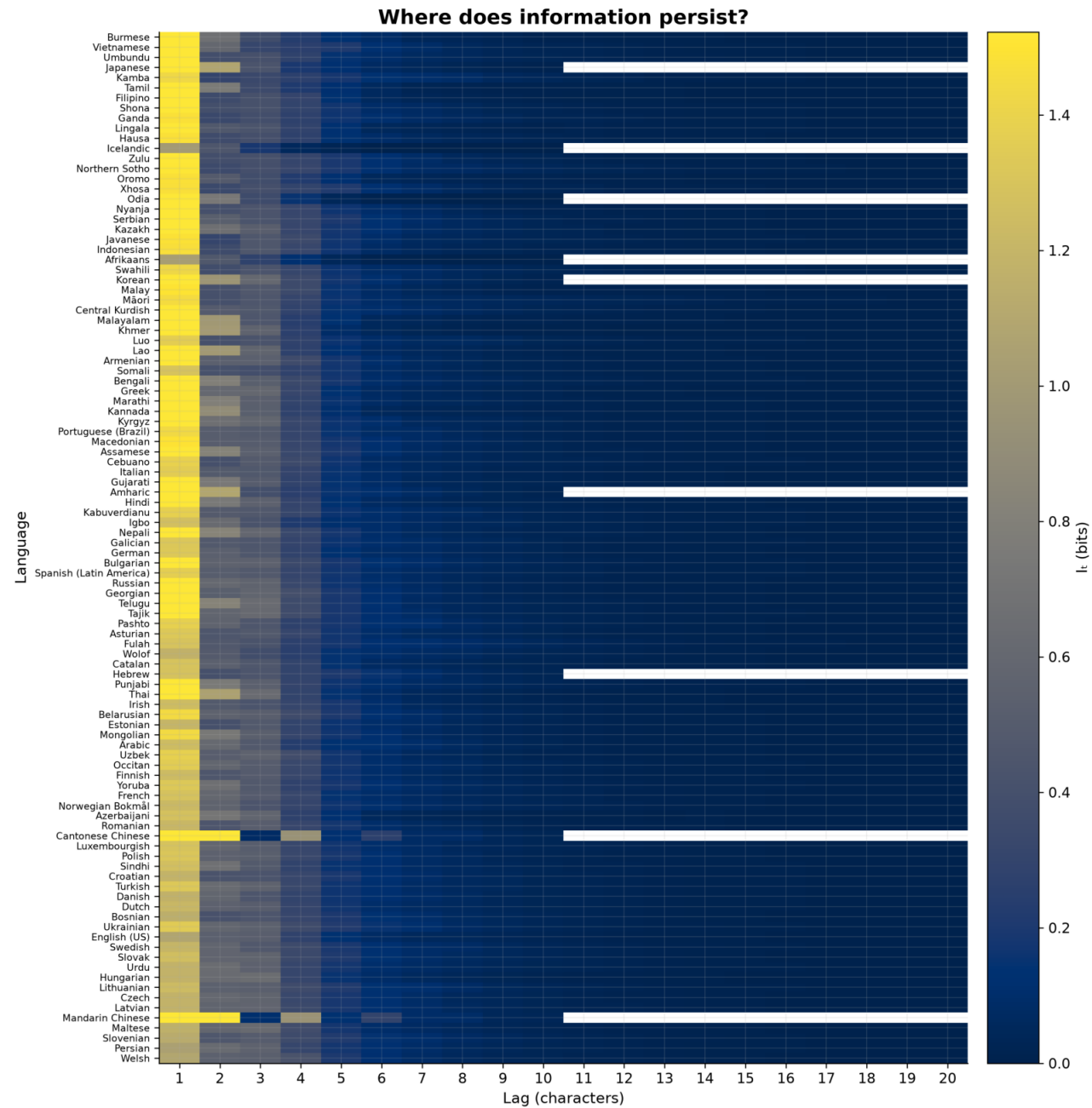


Caption — Each faint line is one language's locality curve; the mean, median and interquartile band summarize the cross-language distribution.

## Information locality after speech-time calibration

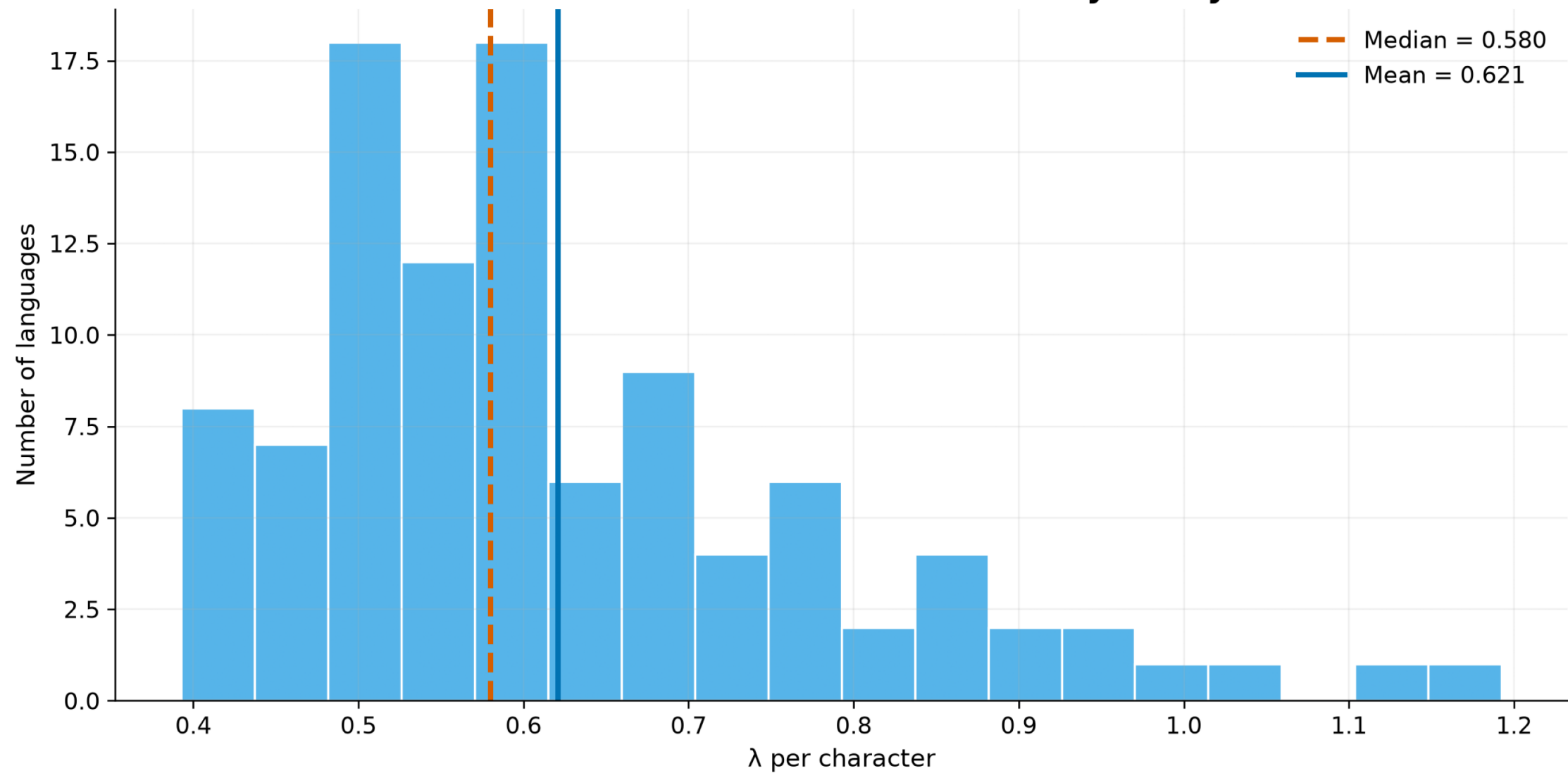


Caption — The same locality curves in seconds, showing how speech-rate calibration changes temporal persistence relative to character persistence.



Caption — A language-by-lag heatmap that makes the complete 102-language locality surface visible without relying on 102 overlapping line colours.

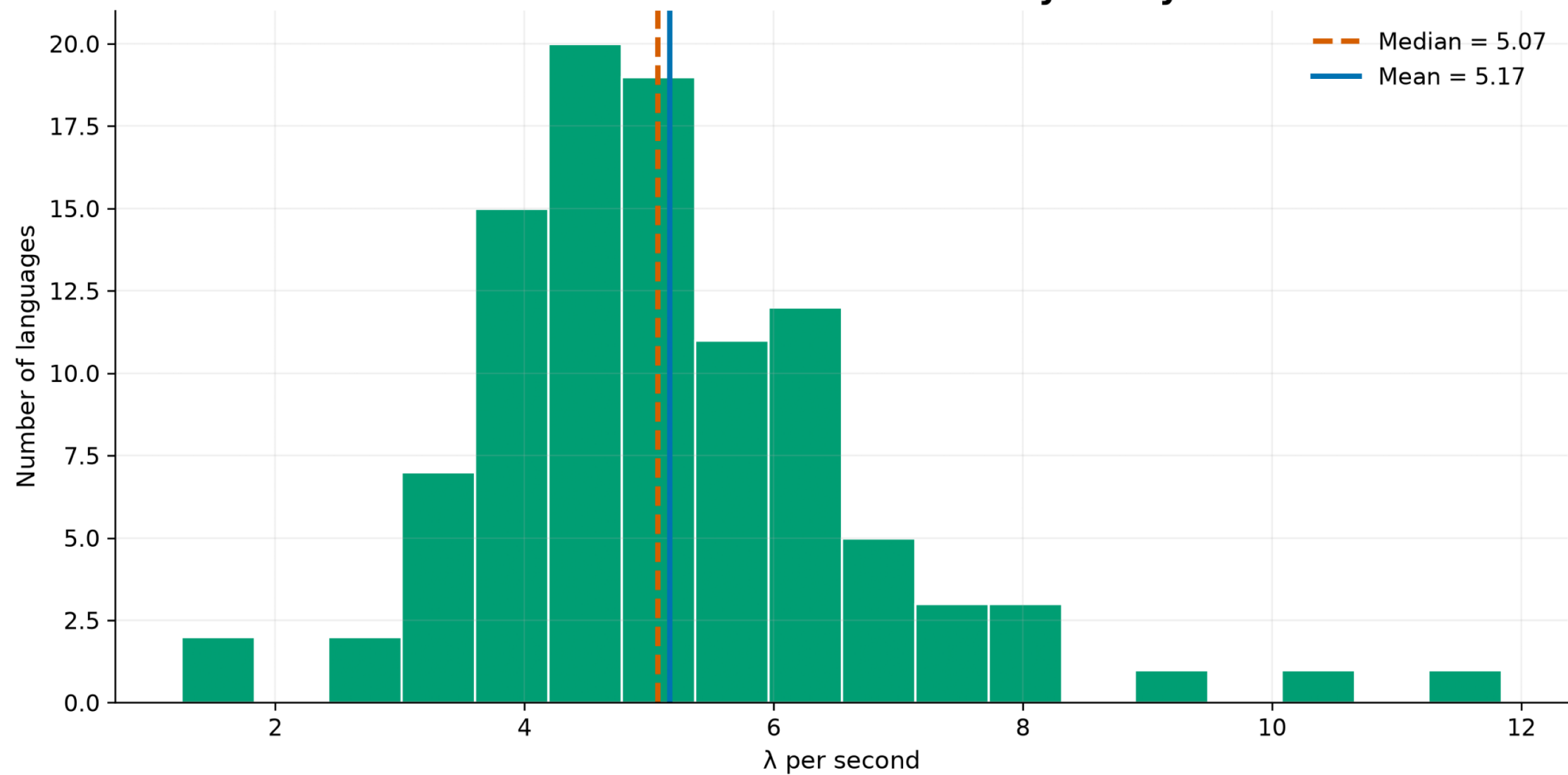
## Distribution of character-domain locality decay rates



Caption — The language-level distribution of the exponential decay parameter in character space.

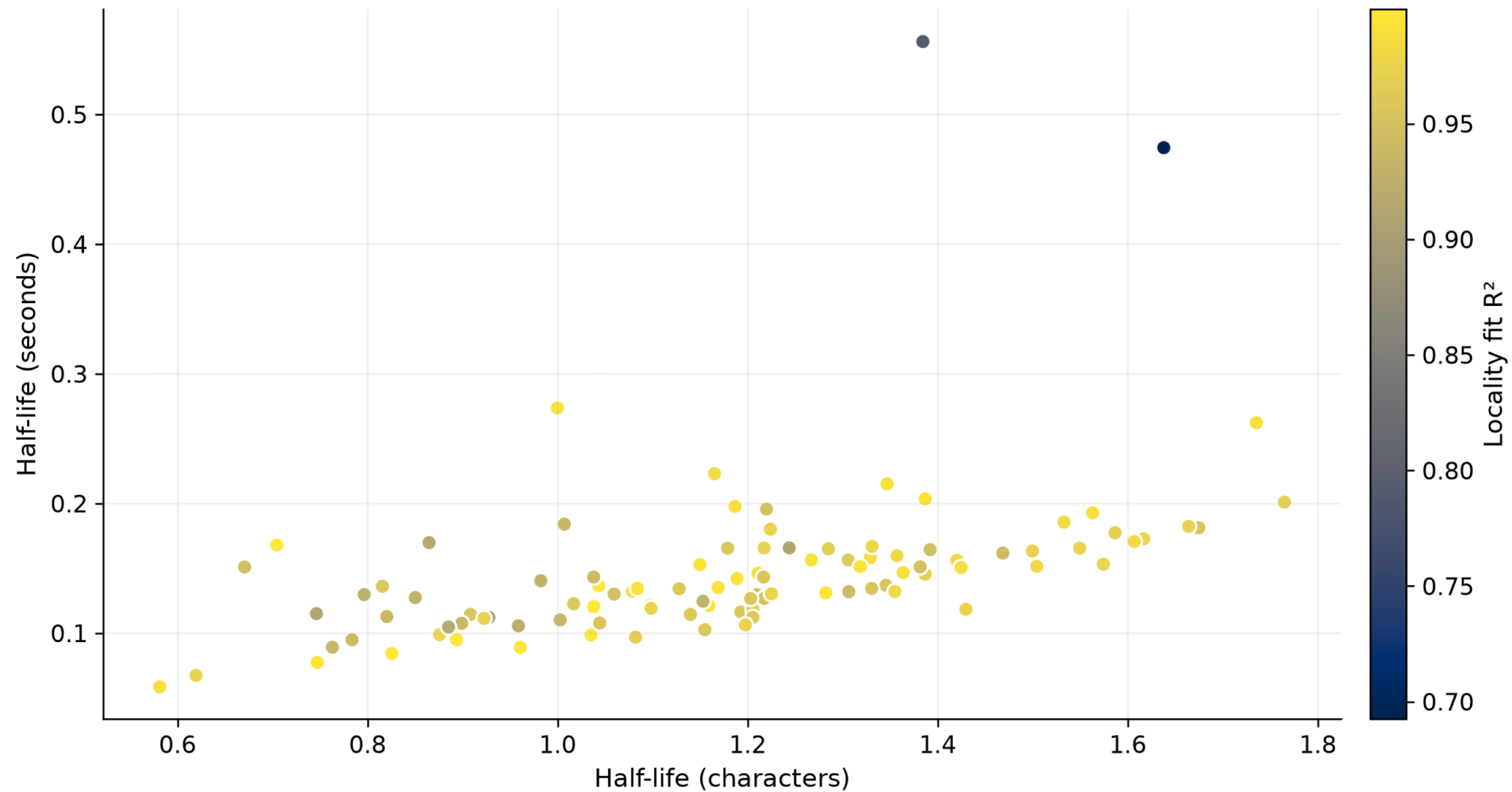


## Distribution of time-domain locality decay rates



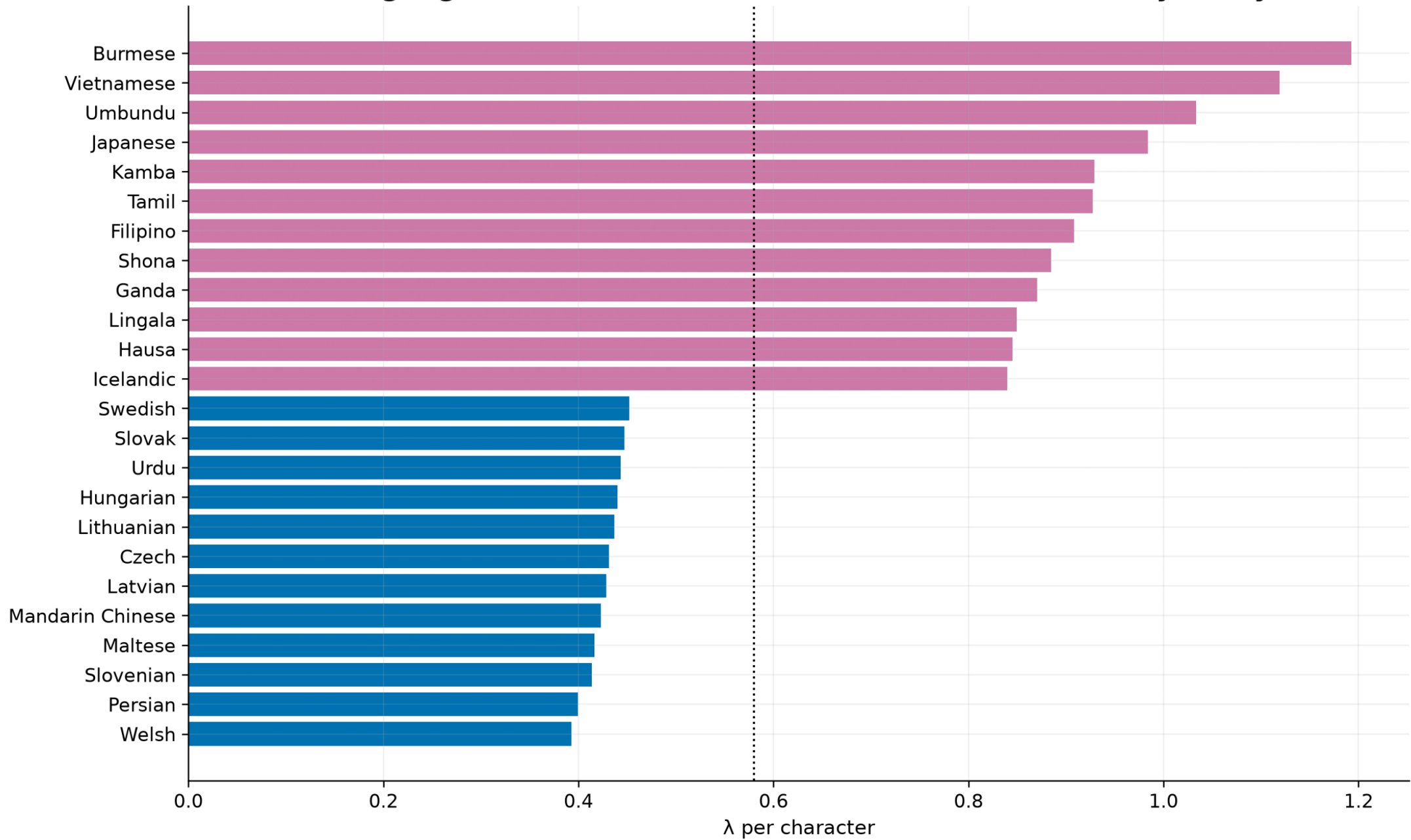
Caption — The language-level distribution of the exponential decay parameter after speech-time calibration.

## Information half-life



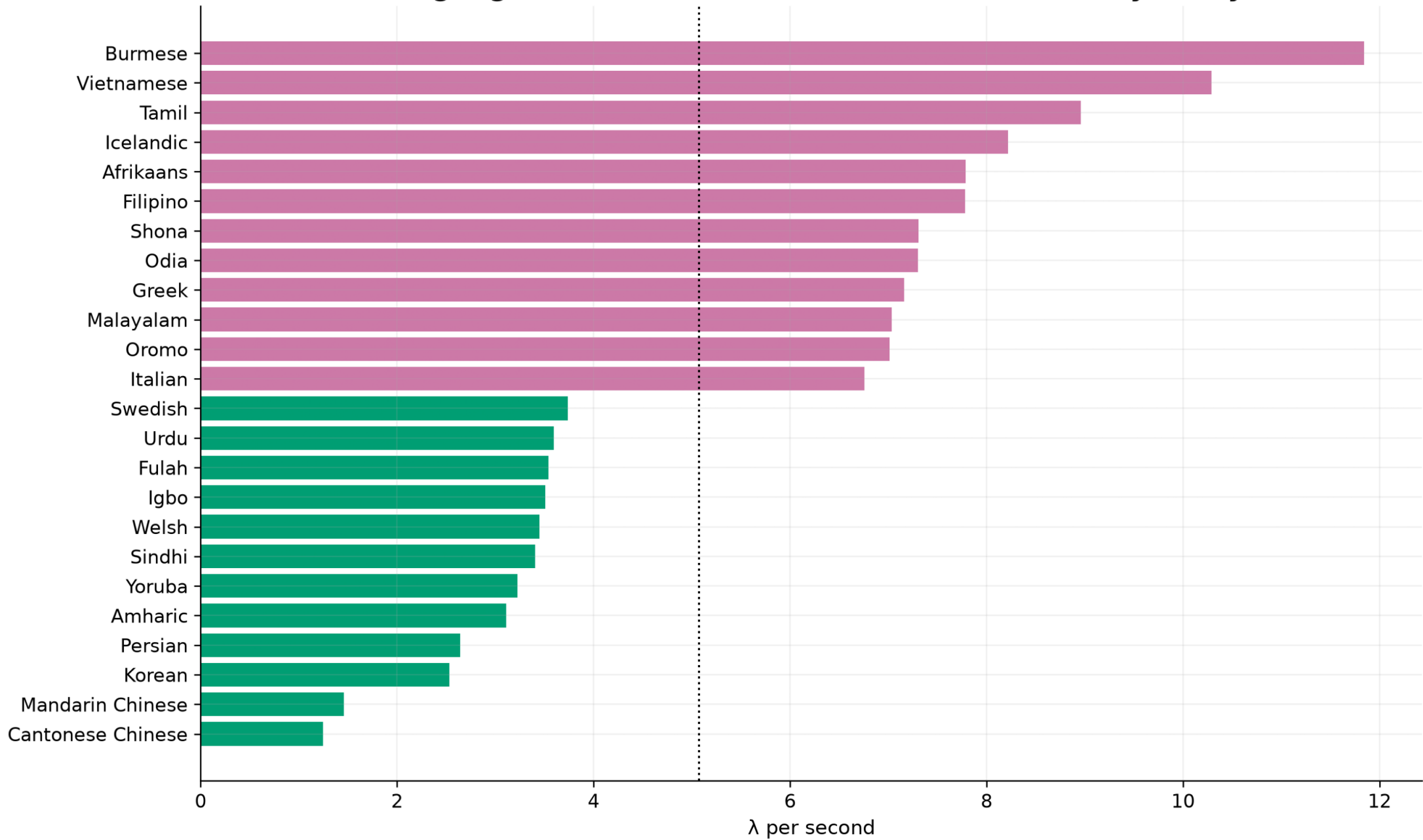
Caption — Half-life is  $\ln(2)/\lambda$  and translates the fitted decay parameter into an interpretable persistence distance.

# Languages at the two ends of character-domain locality decay



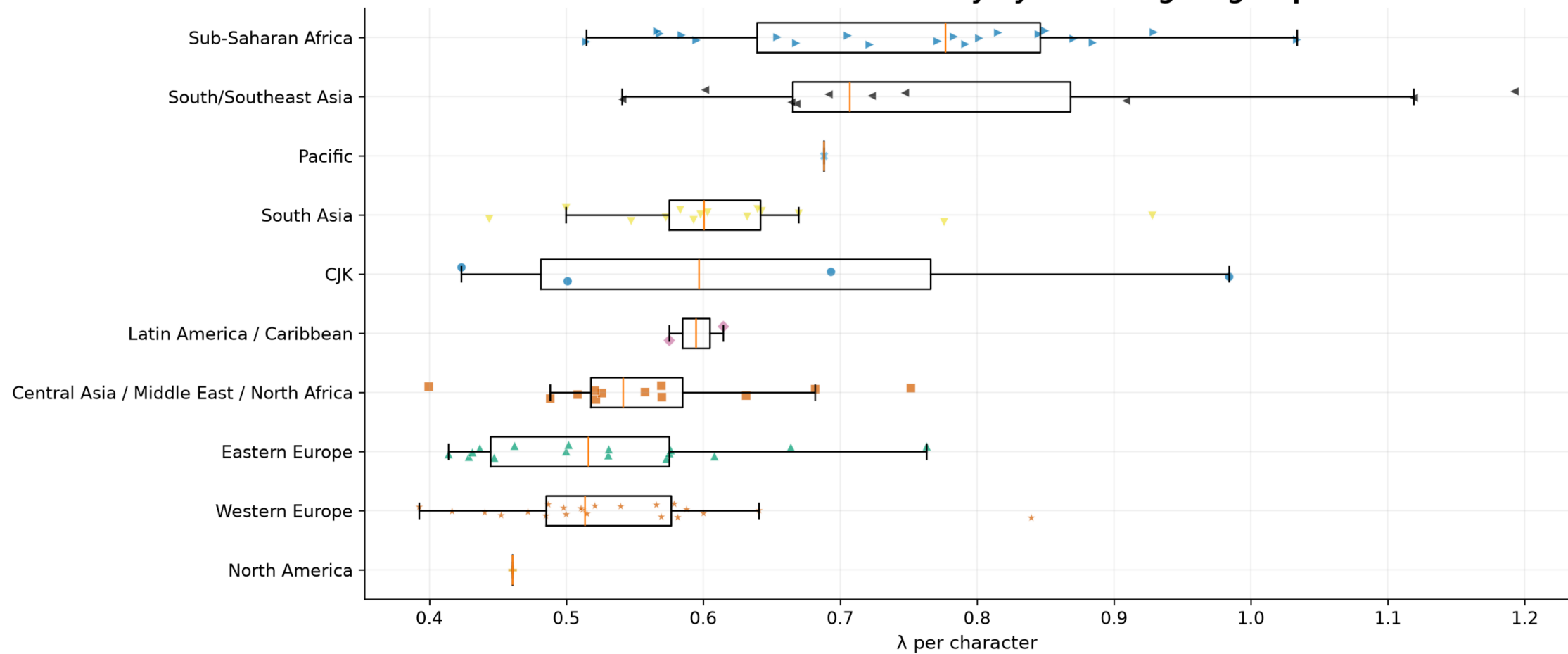
Caption — Descriptive extremum plot showing the 12 lowest- and 12 highest-rate languages; it is not an overall ranking of languages.

# Languages at the two ends of time-domain locality decay



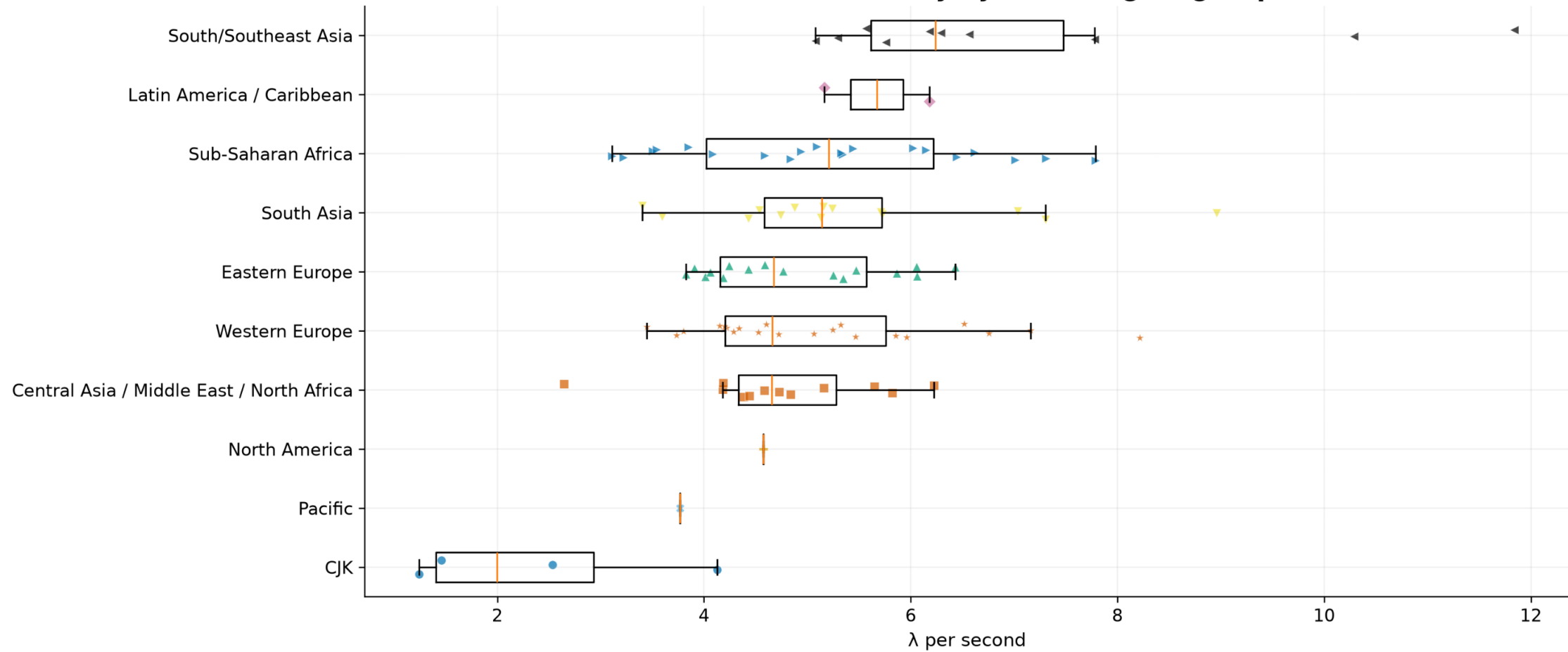
Caption — Descriptive extremum plot showing the 12 lowest- and 12 highest-rate languages; it is not an overall ranking of languages.

## Character-domain locality by broad region group



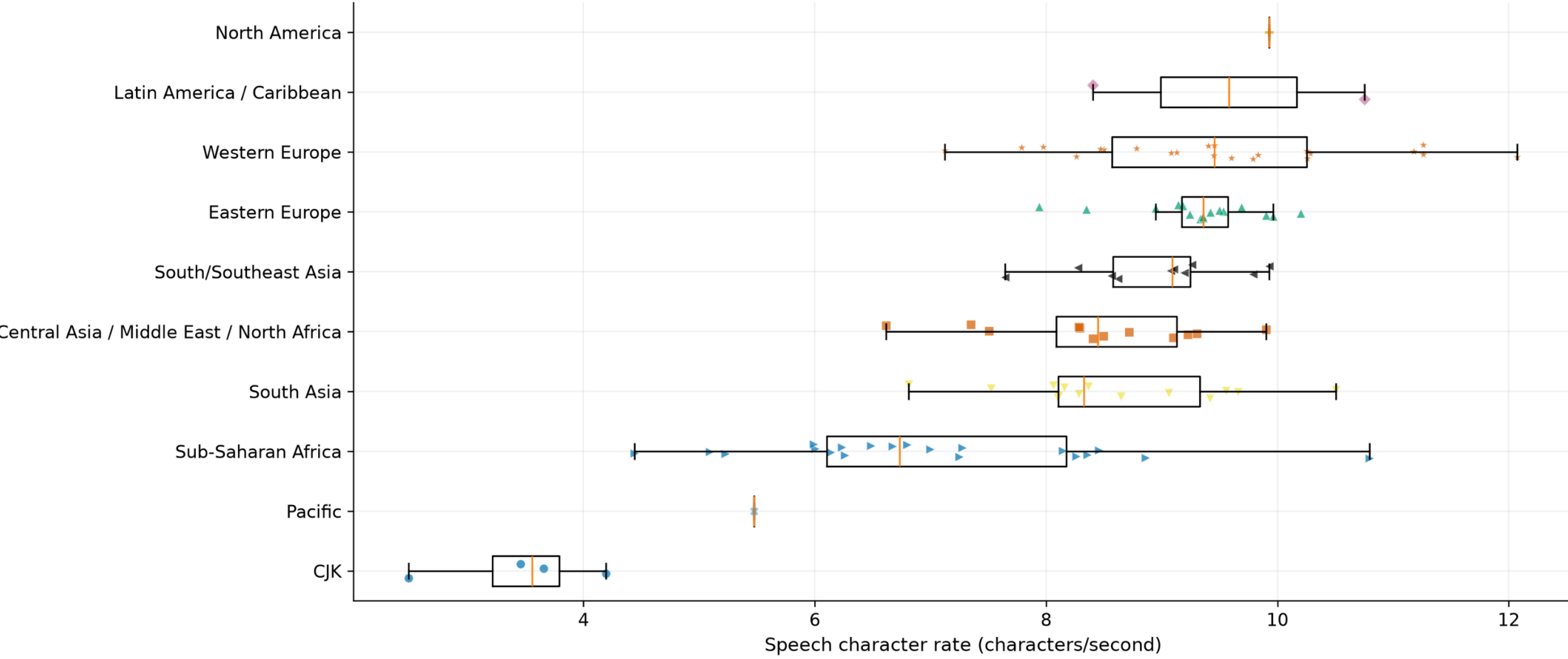
Caption — Boxes summarize within-region distributions and overlaid points show individual languages; sparse groups should be interpreted descriptively.

## Time-domain locality by broad region group



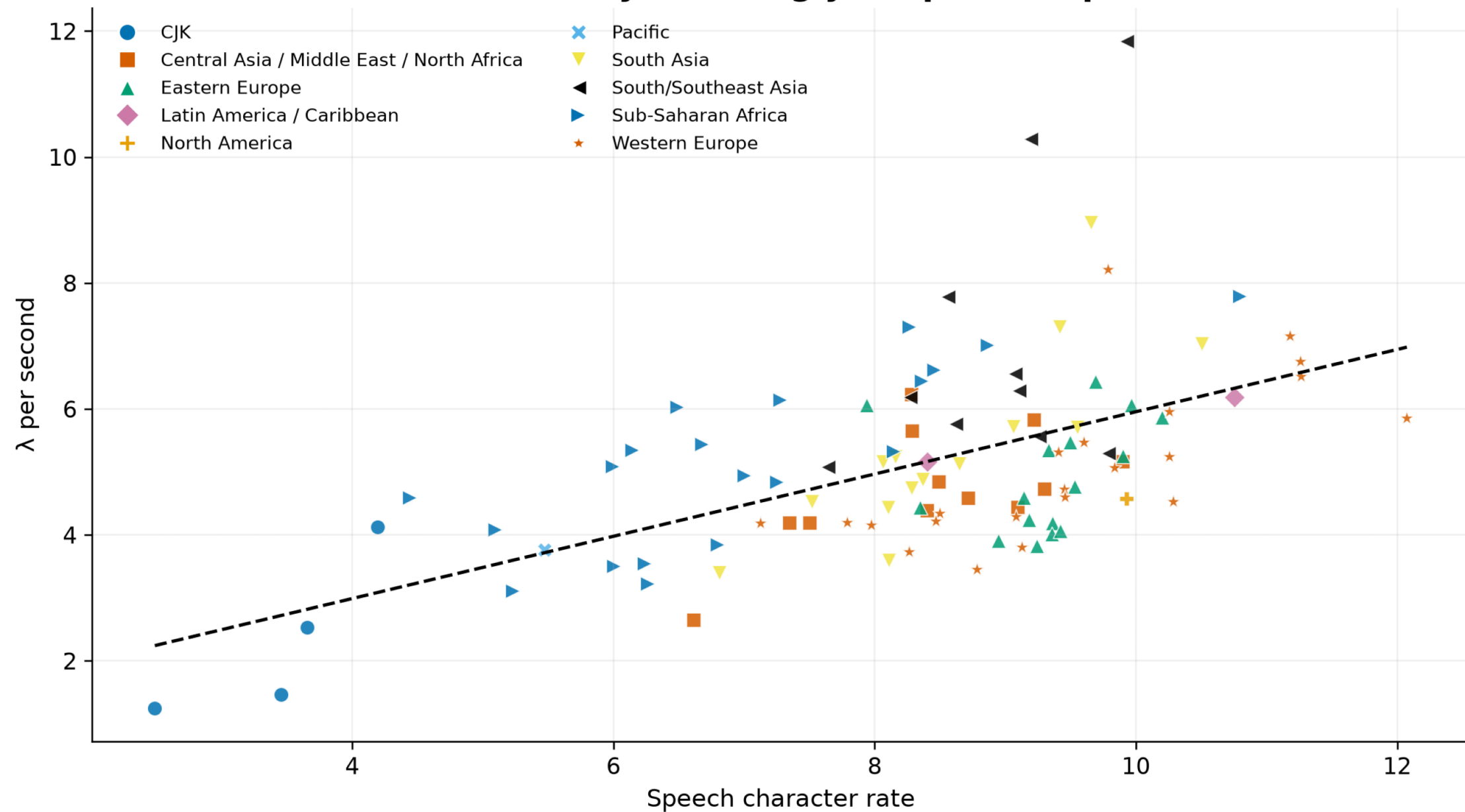
Caption — Boxes summarize within-region distributions and overlaid points show individual languages; sparse groups should be interpreted descriptively.

Speech-rate calibration by broad region group



Caption — Boxes summarize within-region distributions and overlaid points show individual languages; sparse groups should be interpreted descriptively.

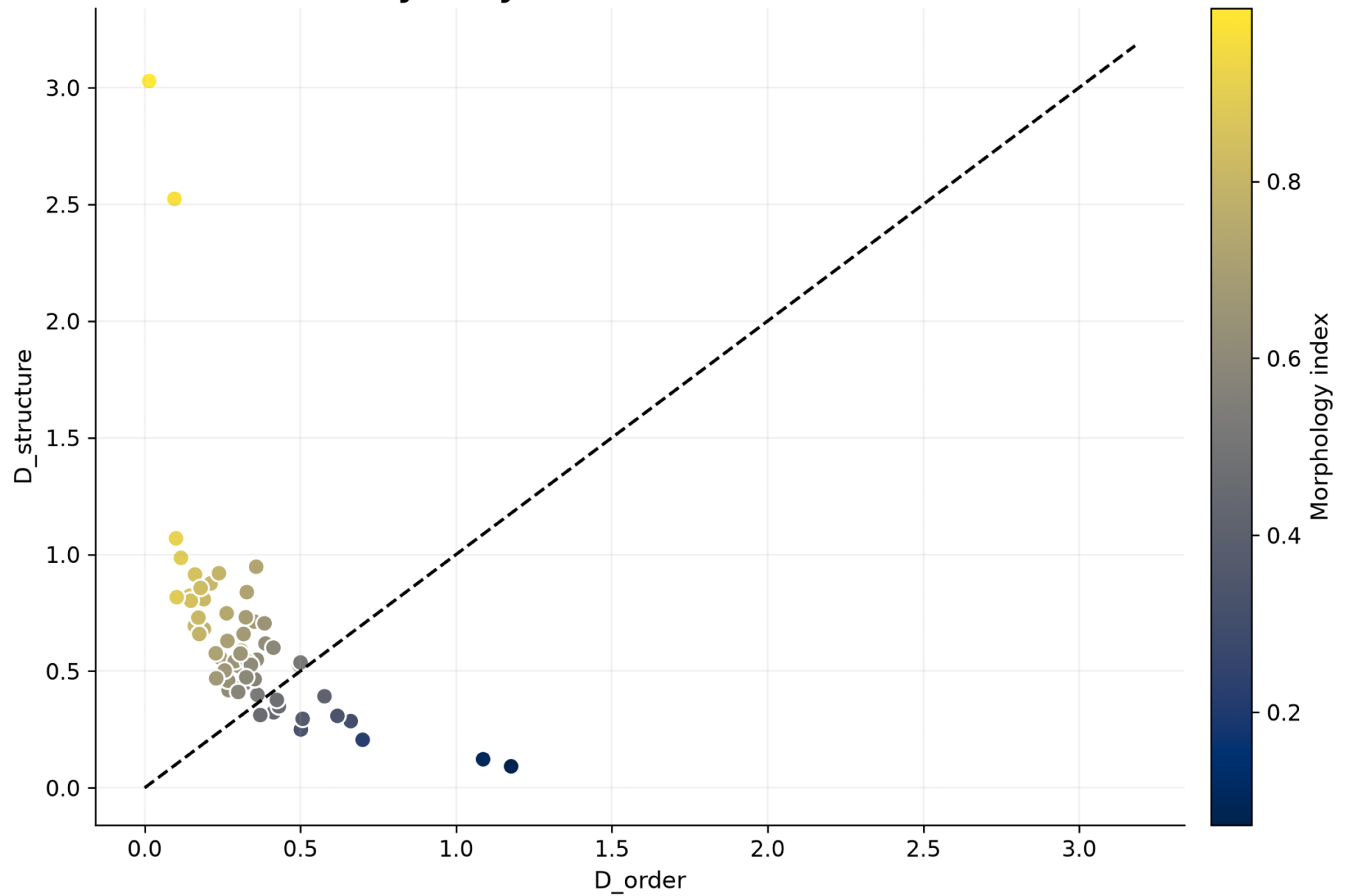
# Time-domain decay is strongly coupled to speech rate



Caption — The positive association between speech-character rate and  $\lambda$ /second is partly algebraically induced because  $\lambda/\text{second} = \lambda/\text{character} \times \text{speech-character-rate}$ .

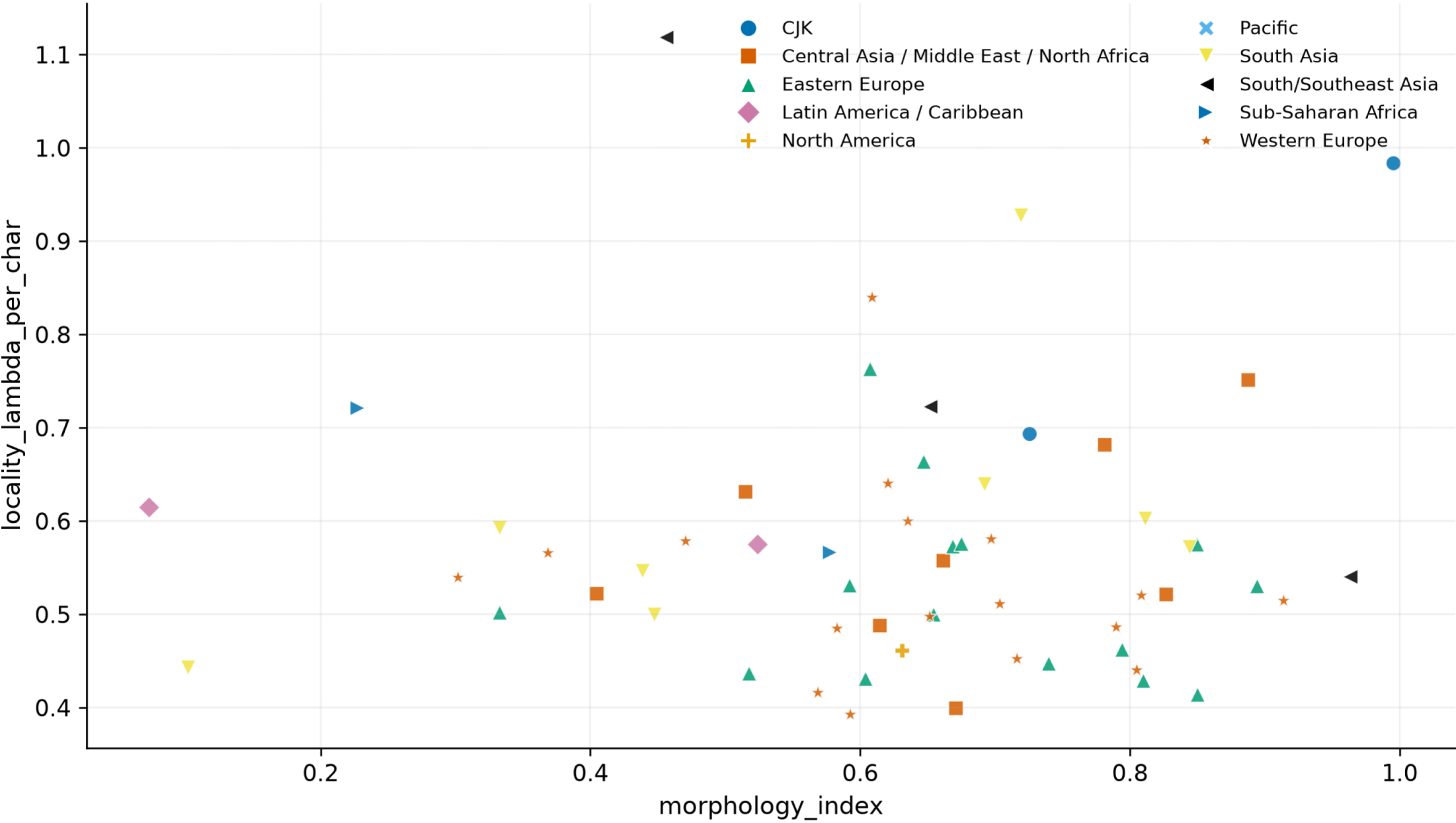


# The analytic-synthetic information trade-off



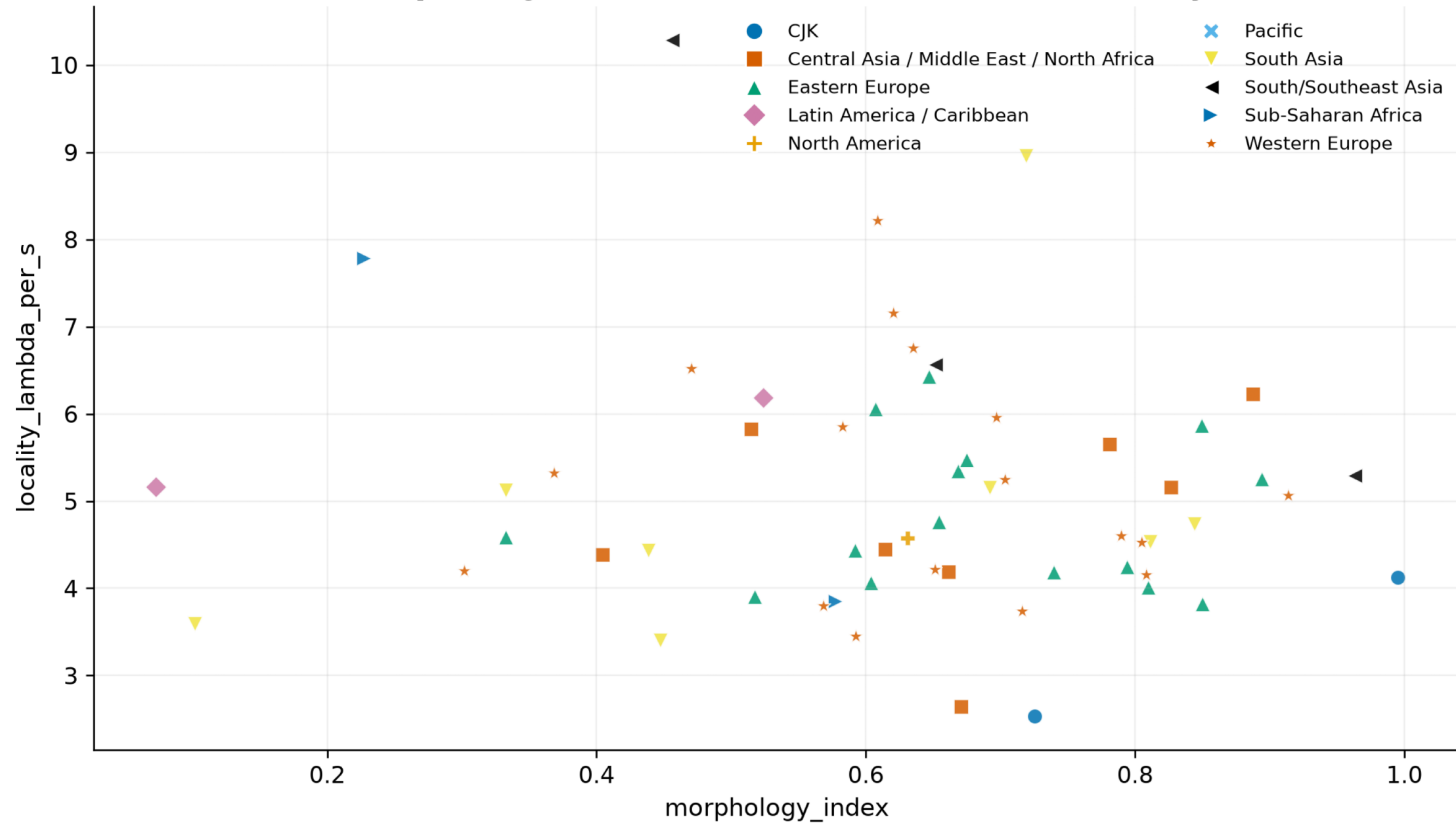
Caption —  $D_{\text{order}}$  and  $D_{\text{structure}}$  show the relative contributions of word order and within-word structure; morphology index is the structural share of their sum.

# Morphological balance vs character-domain locality



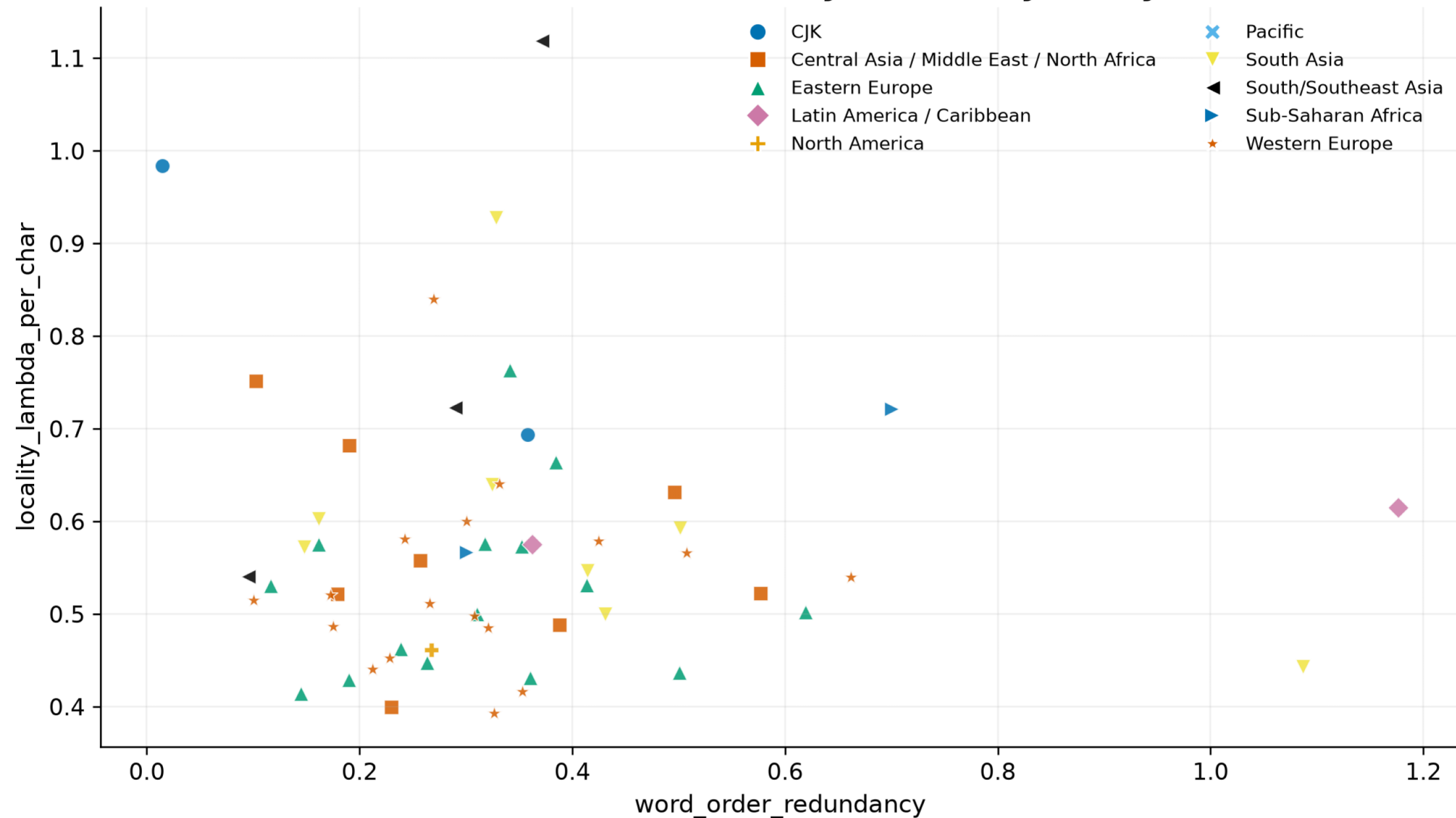
Caption — Bivariate diagnostic for morphology\_index versus locality\_lambda\_per\_char. The plot is descriptive and should not be interpreted causally; it checks whether a simple language-level covariate visibly tracks locality.

# Morphological balance vs time-domain locality



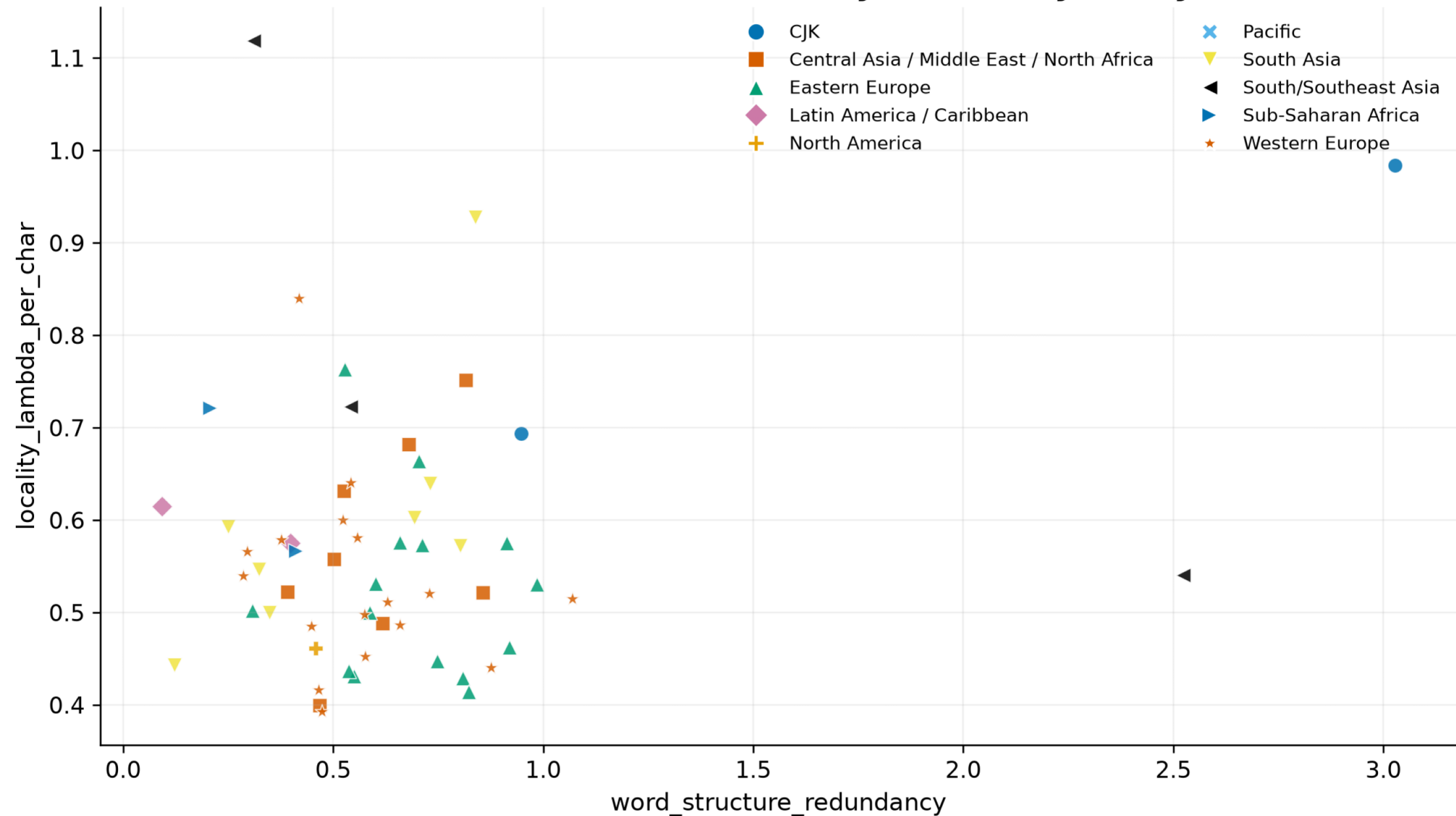
Caption — Bivariate diagnostic for `morphology_index` versus `locality_lambda_per_s`. The plot is descriptive and should not be interpreted causally; it checks whether a simple language-level covariate visibly tracks locality.

# Word-order redundancy vs locality decay



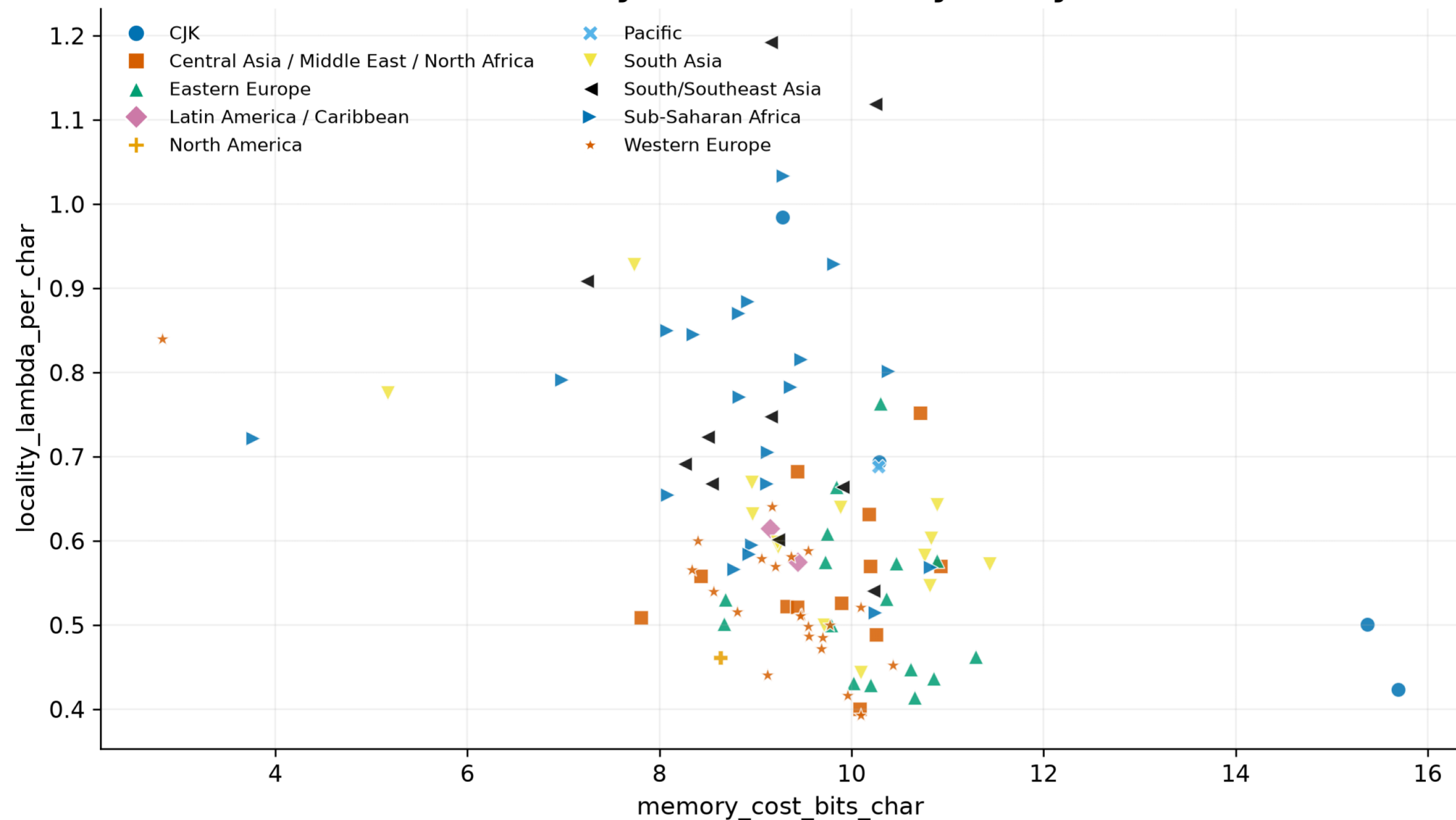
Caption — Bivariate diagnostic for word\_order\_redundancy versus locality\_lambda\_per\_char. The plot is descriptive and should not be interpreted causally; it checks whether a simple language-level covariate visibly tracks locality.

# Word-structure redundancy vs locality decay



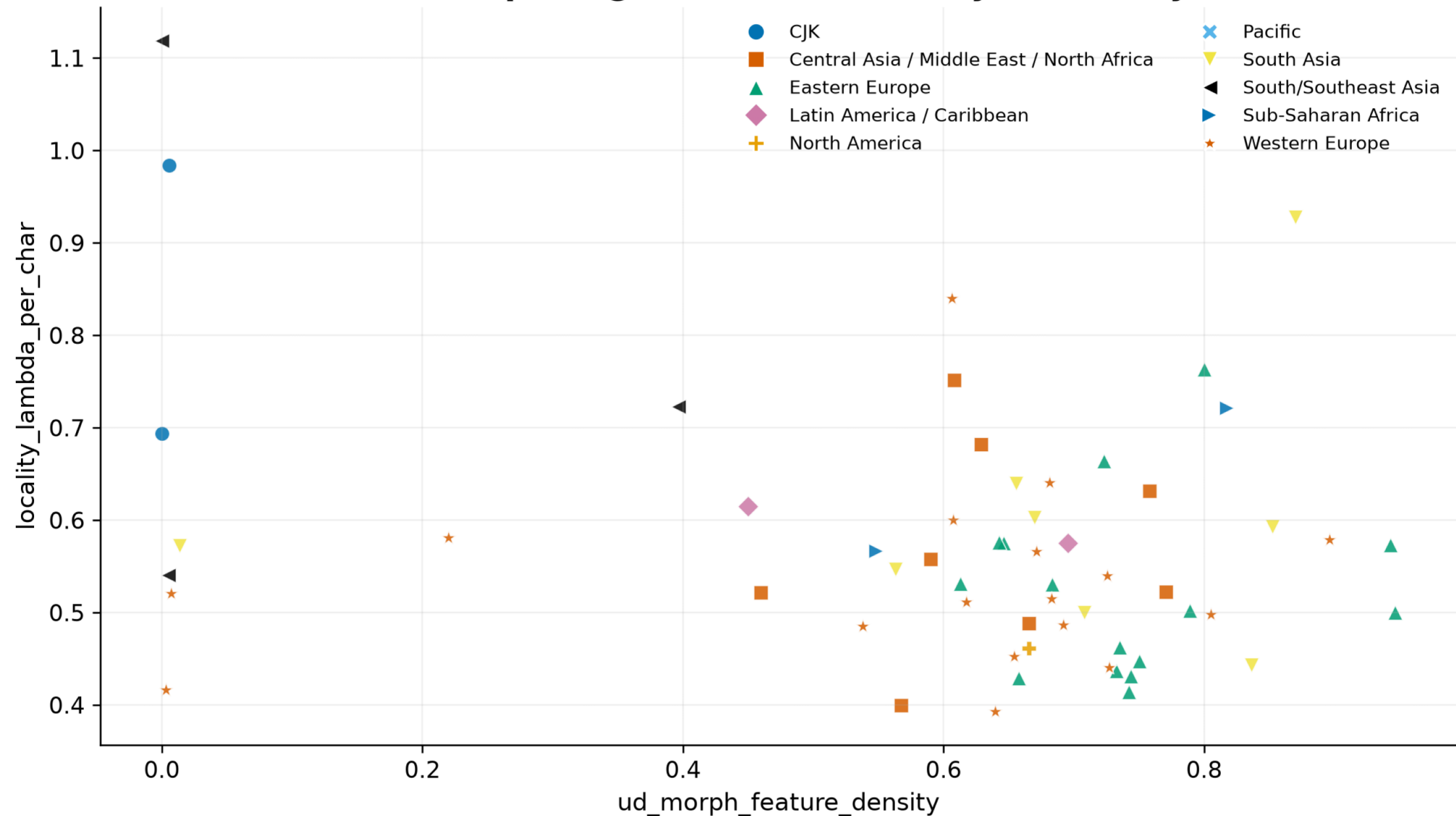
Caption — Bivariate diagnostic for word\_structure\_redundancy versus locality\_lambda\_per\_char. The plot is descriptive and should not be interpreted causally; it checks whether a simple language-level covariate visibly tracks locality.

# Memory cost and locality decay



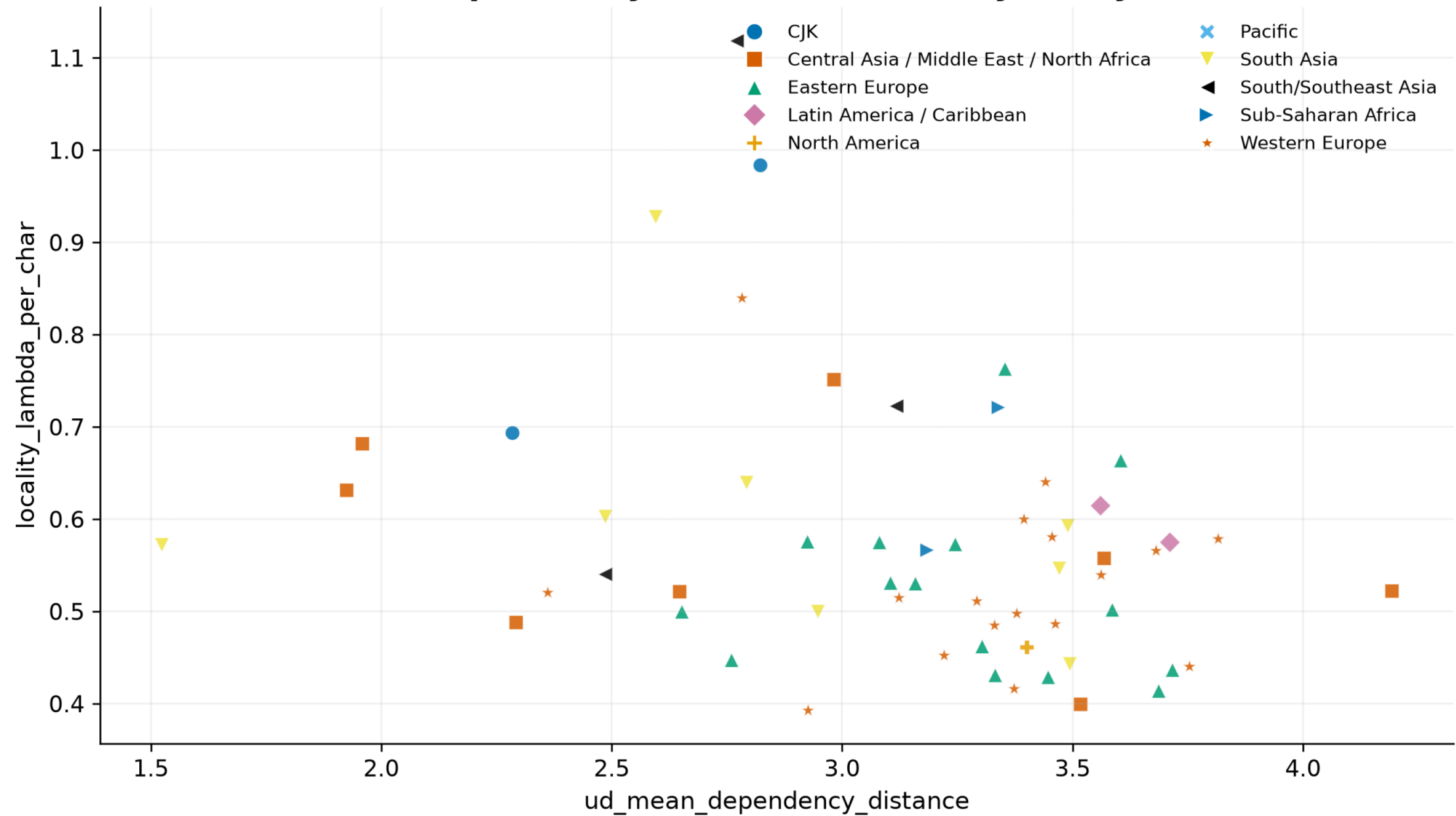
Caption — Bivariate diagnostic for memory\_cost\_bits\_char versus locality\_lambda\_per\_char. The plot is descriptive and should not be interpreted causally; it checks whether a simple language-level covariate visibly tracks locality.

# UD morphological feature density vs locality



Caption — Bivariate diagnostic for ud\_morph\_feature\_density versus locality\_lambda\_per\_char. The plot is descriptive and should not be interpreted causally; it checks whether a simple language-level covariate visibly tracks locality.

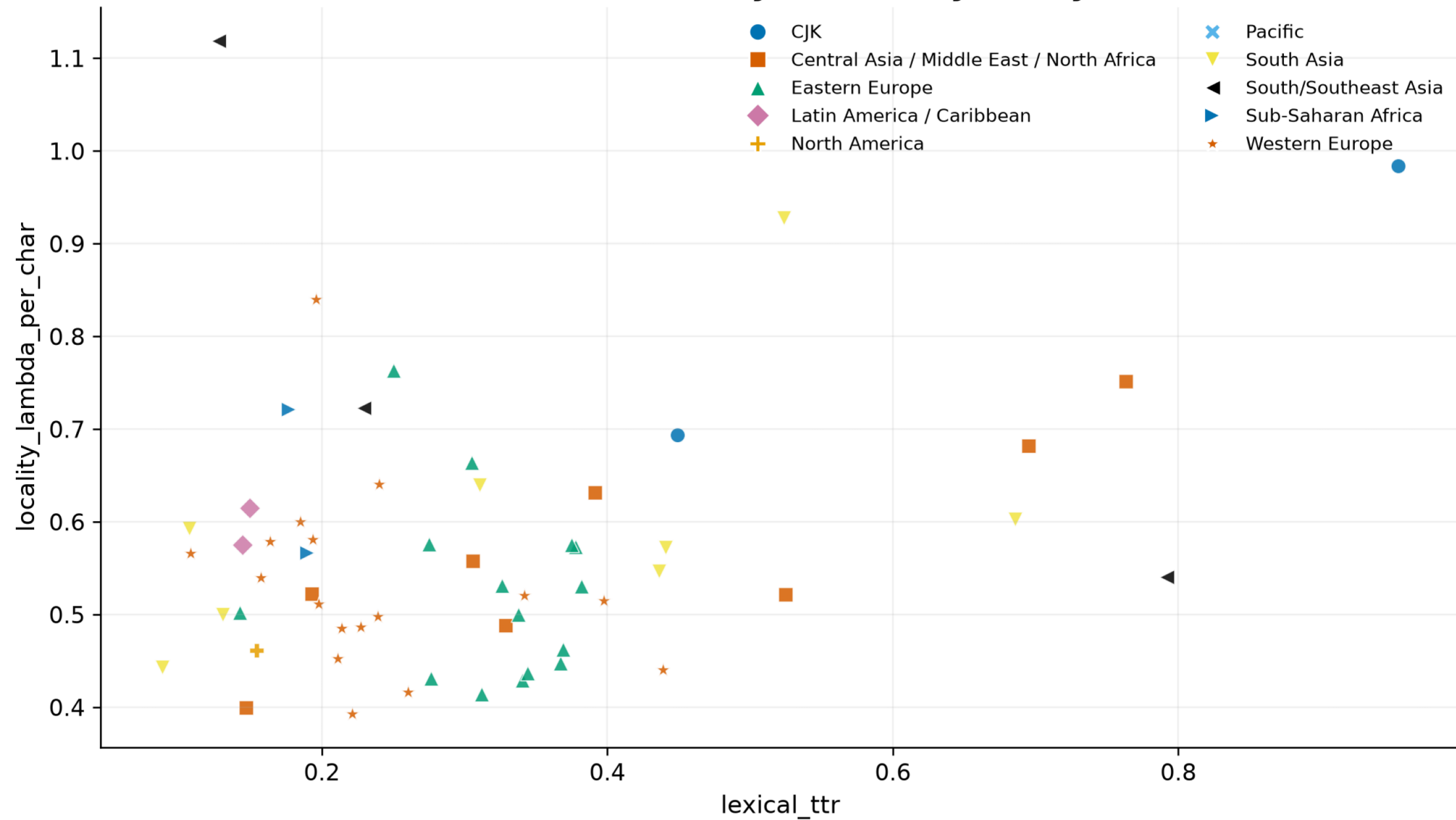
# Dependency distance vs locality decay



Caption — Bivariate diagnostic for `ud_mean_dependency_distance` versus `locality_lambda_per_char`. The plot is descriptive and should not be interpreted causally; it checks whether a simple language-level covariate visibly tracks locality.

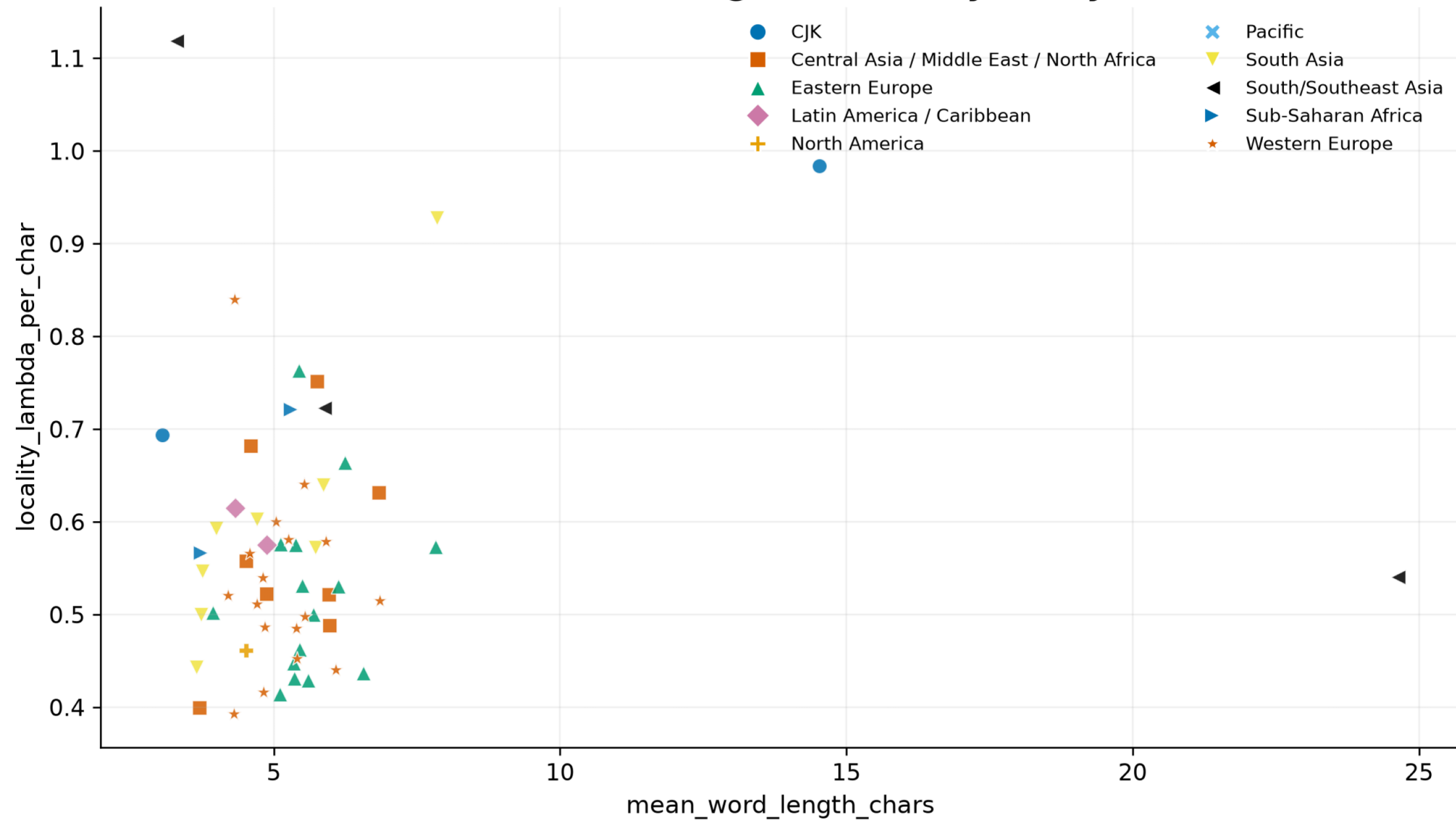


# Lexical diversity vs locality decay



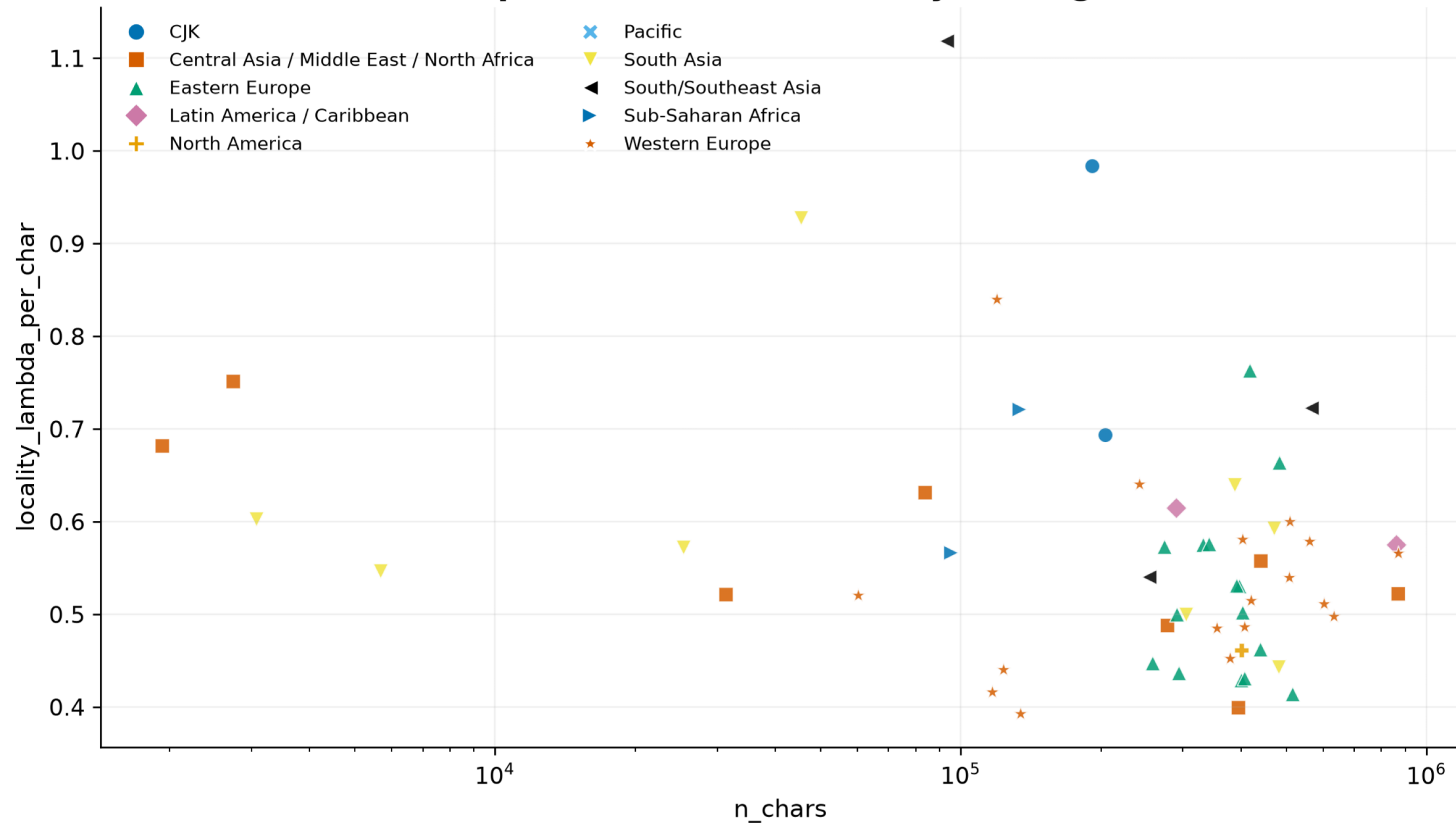
Caption — Bivariate diagnostic for lexical\_ttr versus locality\_lambda\_per\_char. The plot is descriptive and should not be interpreted causally; it checks whether a simple language-level covariate visibly tracks locality.

# Mean word length vs locality decay



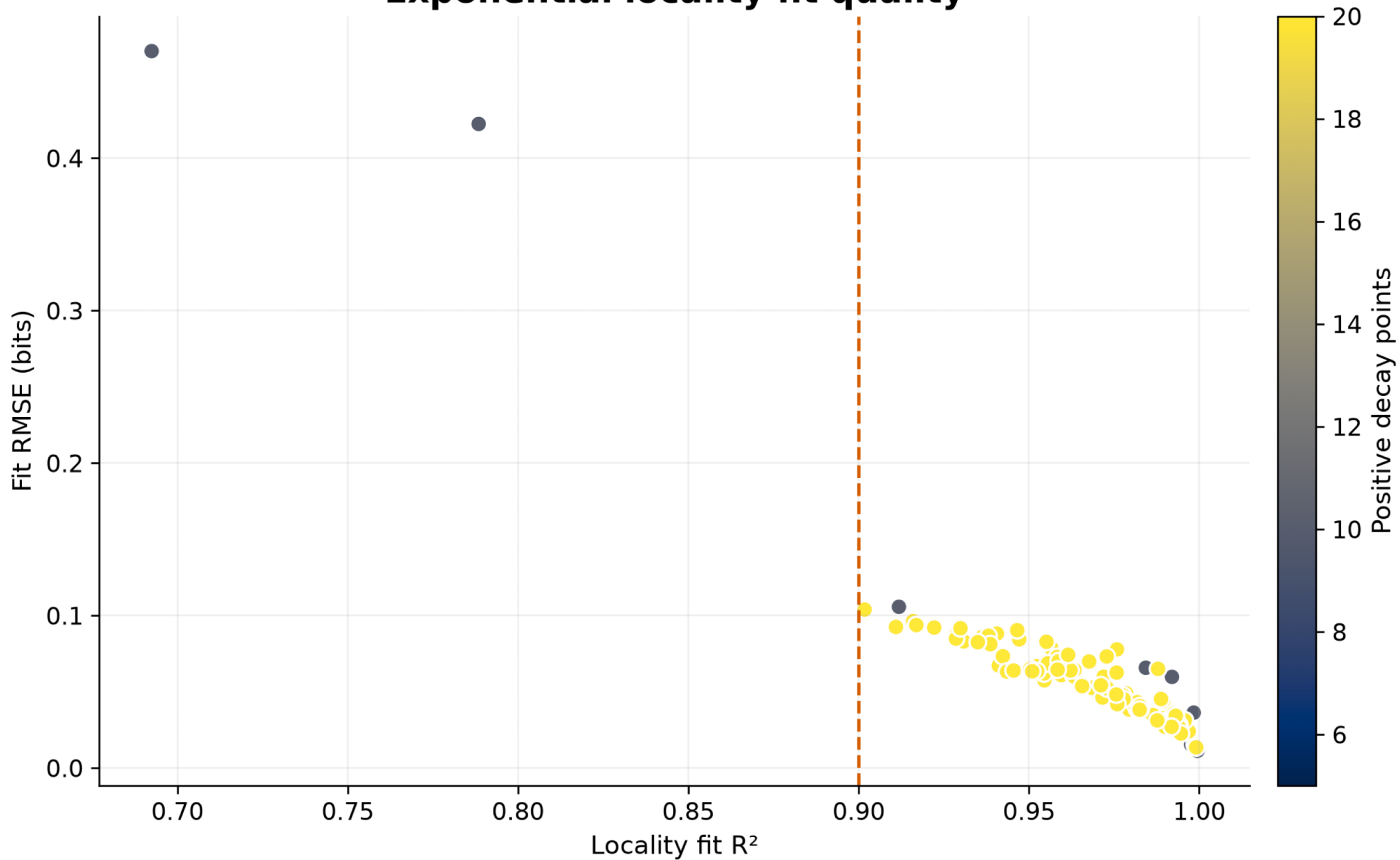
Caption — Bivariate diagnostic for mean\_word\_length\_chars versus locality\_lambda\_per\_char. The plot is descriptive and should not be interpreted causally; it checks whether a simple language-level covariate visibly tracks locality.

## Corpus size is not obviously driving $\lambda$



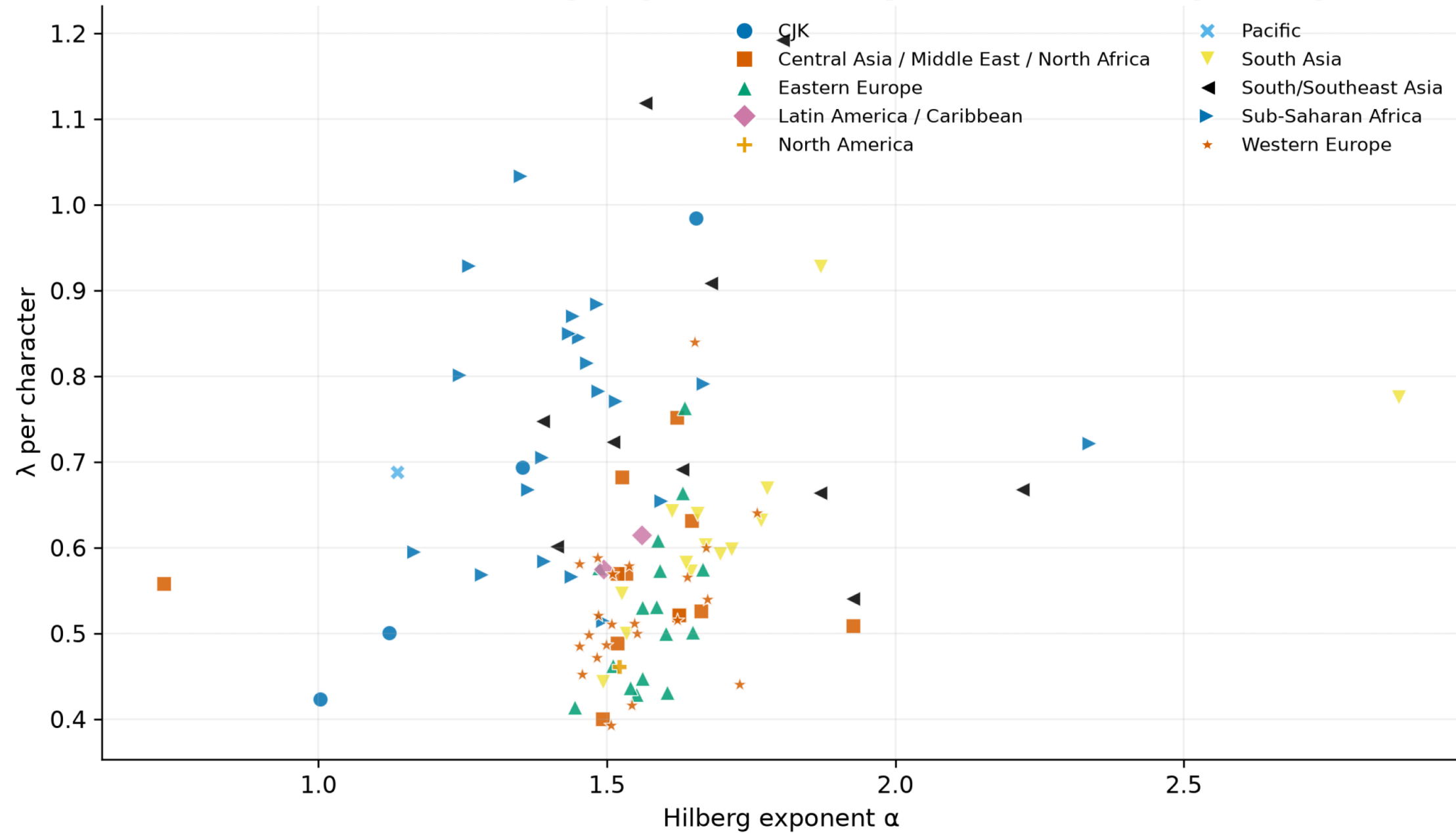
Caption — Bivariate diagnostic for  $n\_chars$  versus locality\_lambda\_per\_char. The plot is descriptive and should not be interpreted causally; it checks whether a simple language-level covariate visibly tracks locality.

# Exponential locality fit quality



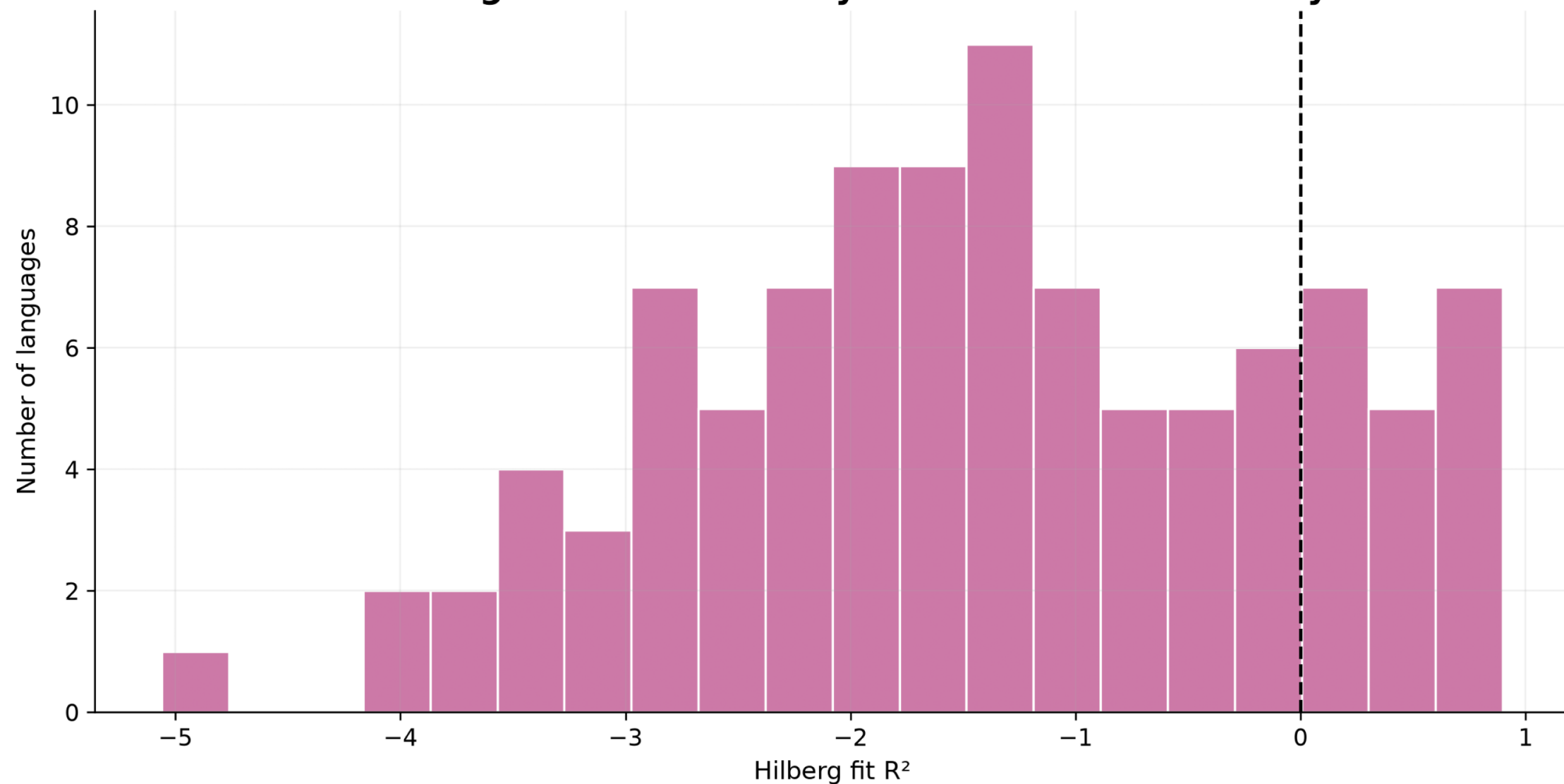
Caption — Most exponential locality fits are strong; the two weakest are Cantonese Chinese and Mandarin Chinese, which merit sensitivity attention.

# Alternative scaling exponent vs exponential locality decay



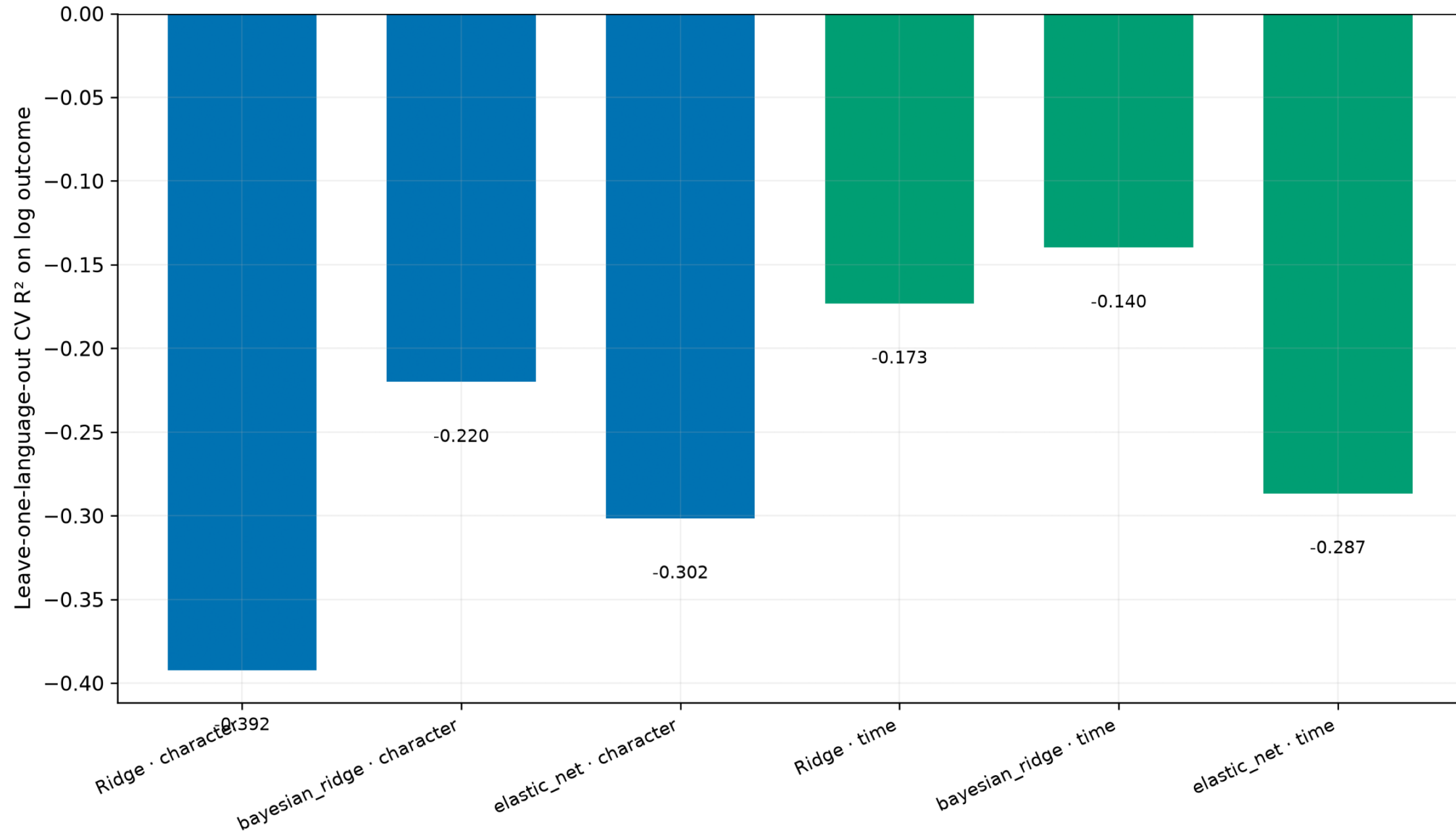
Caption — Hilberg  $\alpha$  and exponential locality  $\lambda$  summarize different aspects of information structure and show only a weak bivariate association in these results.

## The Hilberg fit is substantially weaker than the locality fit



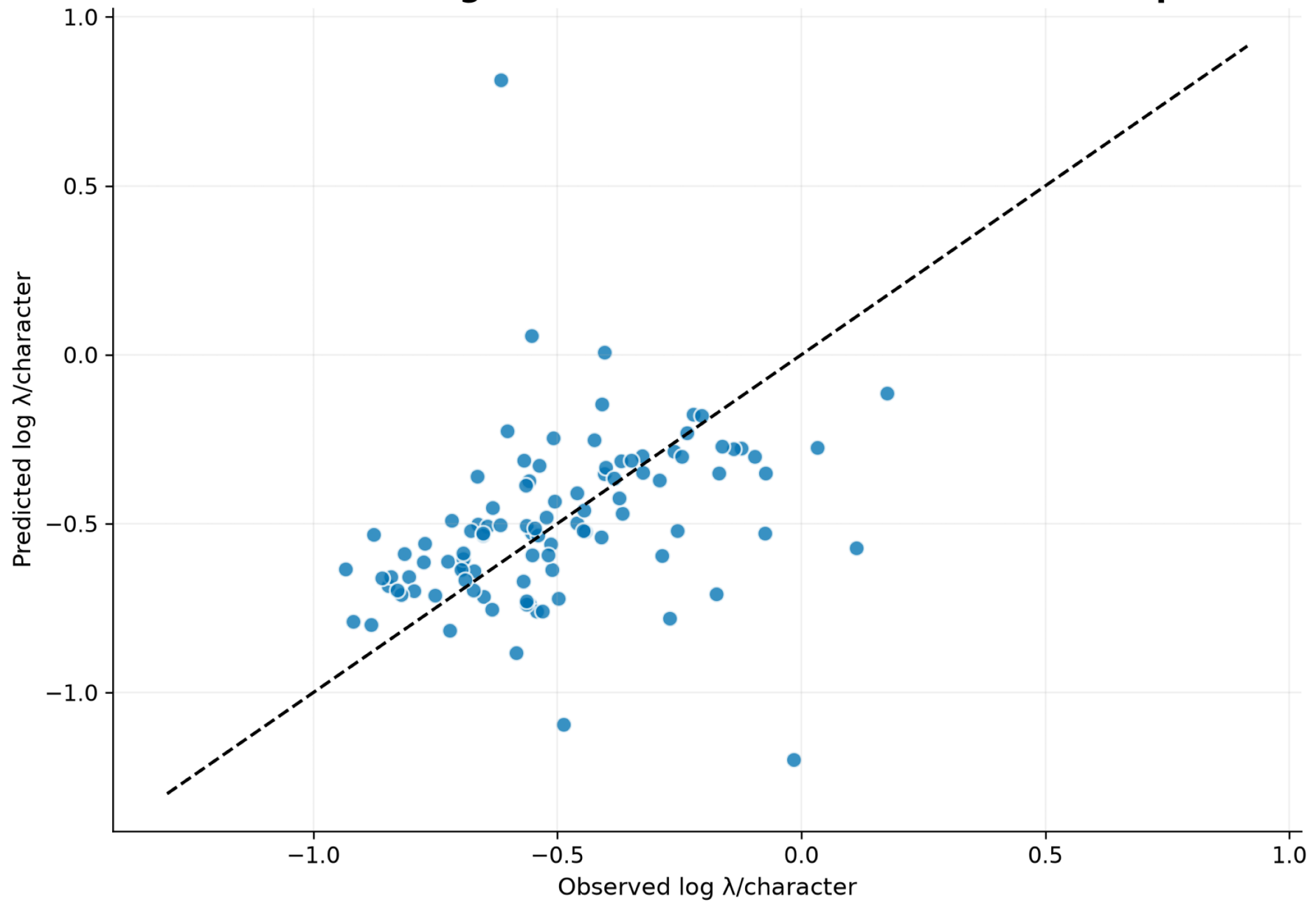
Caption — 83 of 102 Hilberg  $R^2$  values are negative, contrasting with the high  $R^2$  of the primary locality exponential fits.

## Cross-language regression generalization is weak



Caption — All reported cross-validated  $R^2$  values are negative, so the current multivariable feature set does not generalize to held-out languages better than the baseline.

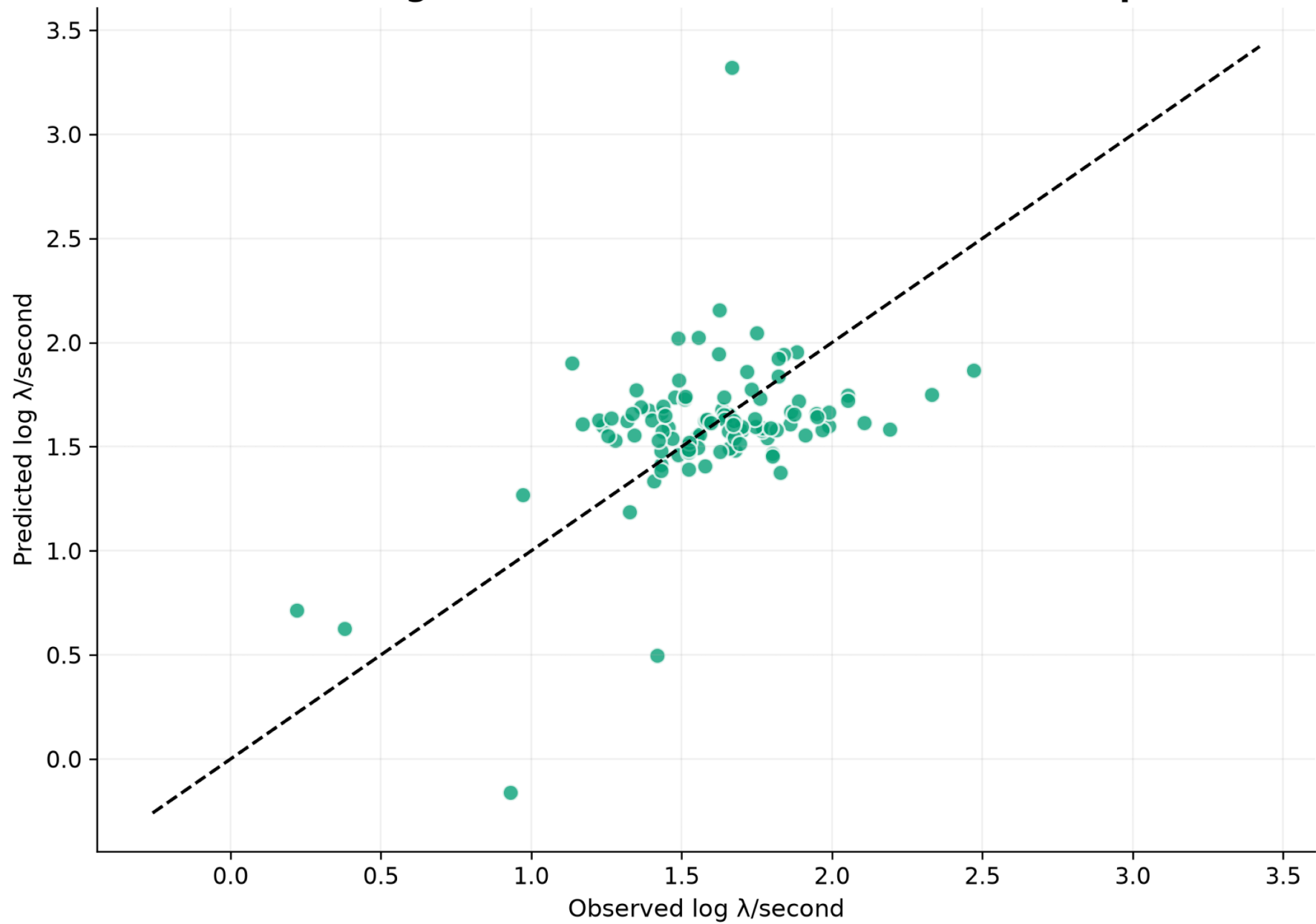
# Character-domain regression: observed vs cross-validated predicted



Caption — Each point is one held-out language and the dashed diagonal is perfect prediction; visible scatter around the diagonal explains the negative cross-validated  $R^2$ .

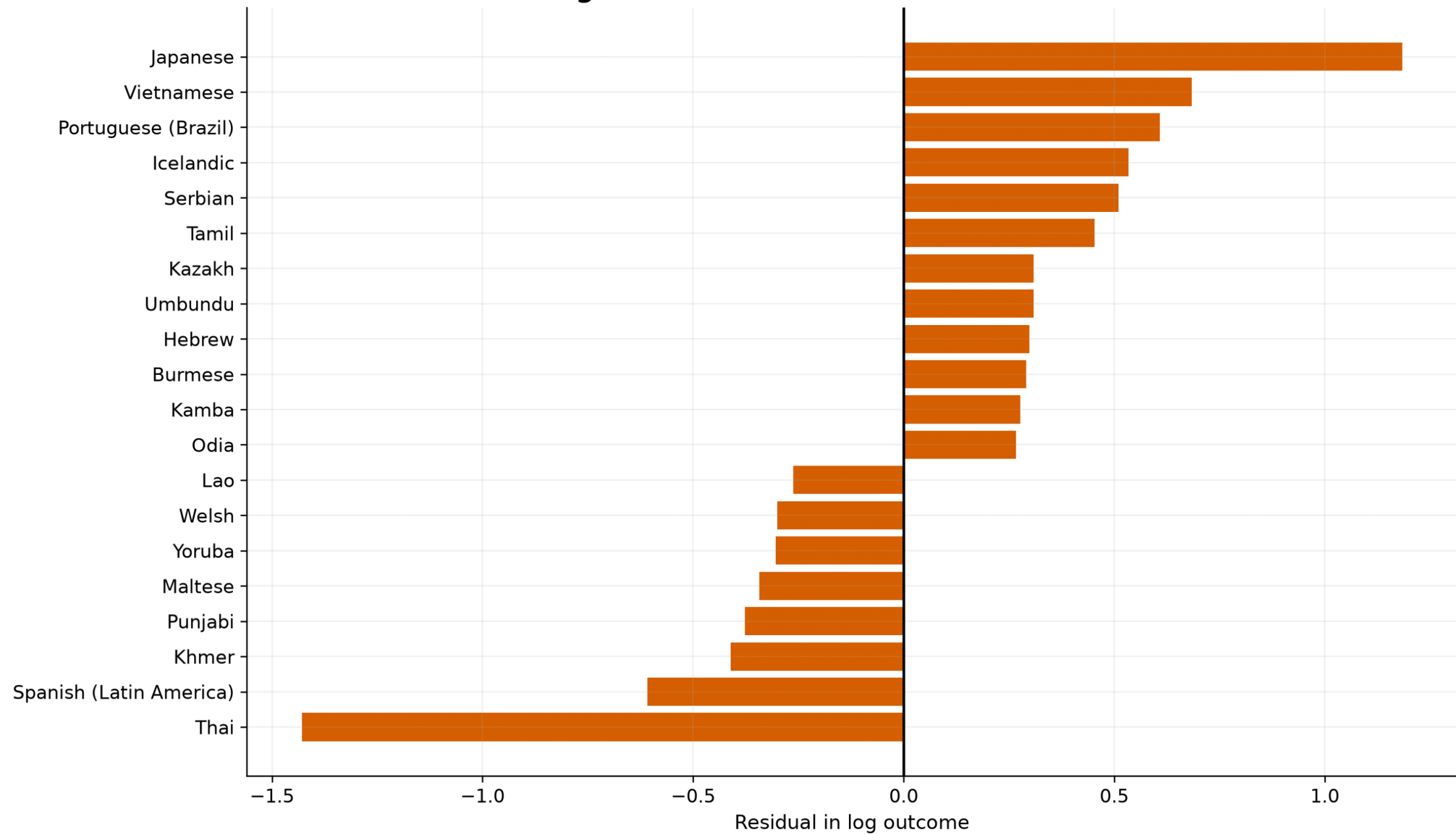


# Time-domain regression: observed vs cross-validated predicted



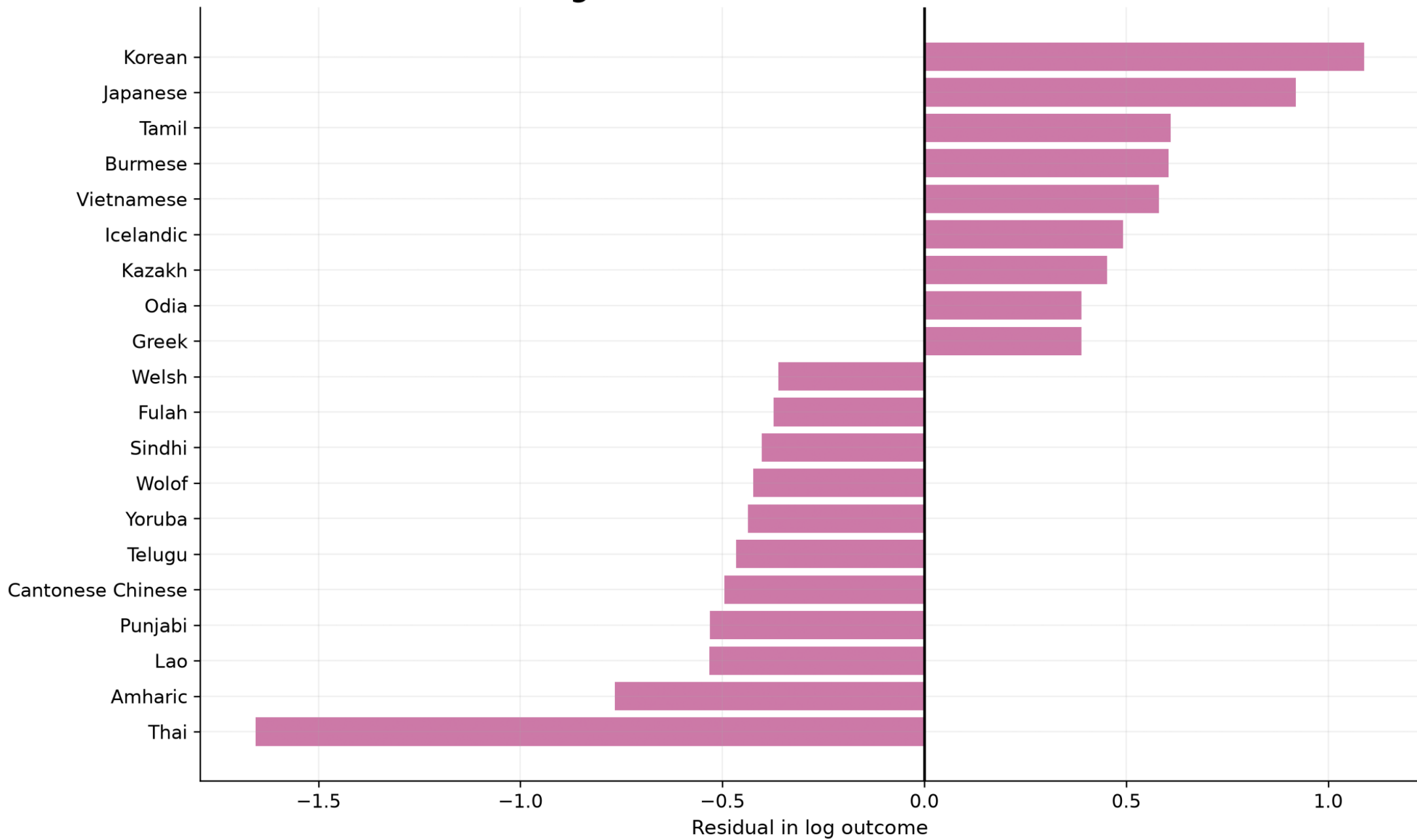
Caption — Each point is one held-out language and the dashed diagonal is perfect prediction; visible scatter around the diagonal explains the negative cross-validated  $R^2$ .

## Largest LOCO-CV errors - character domain



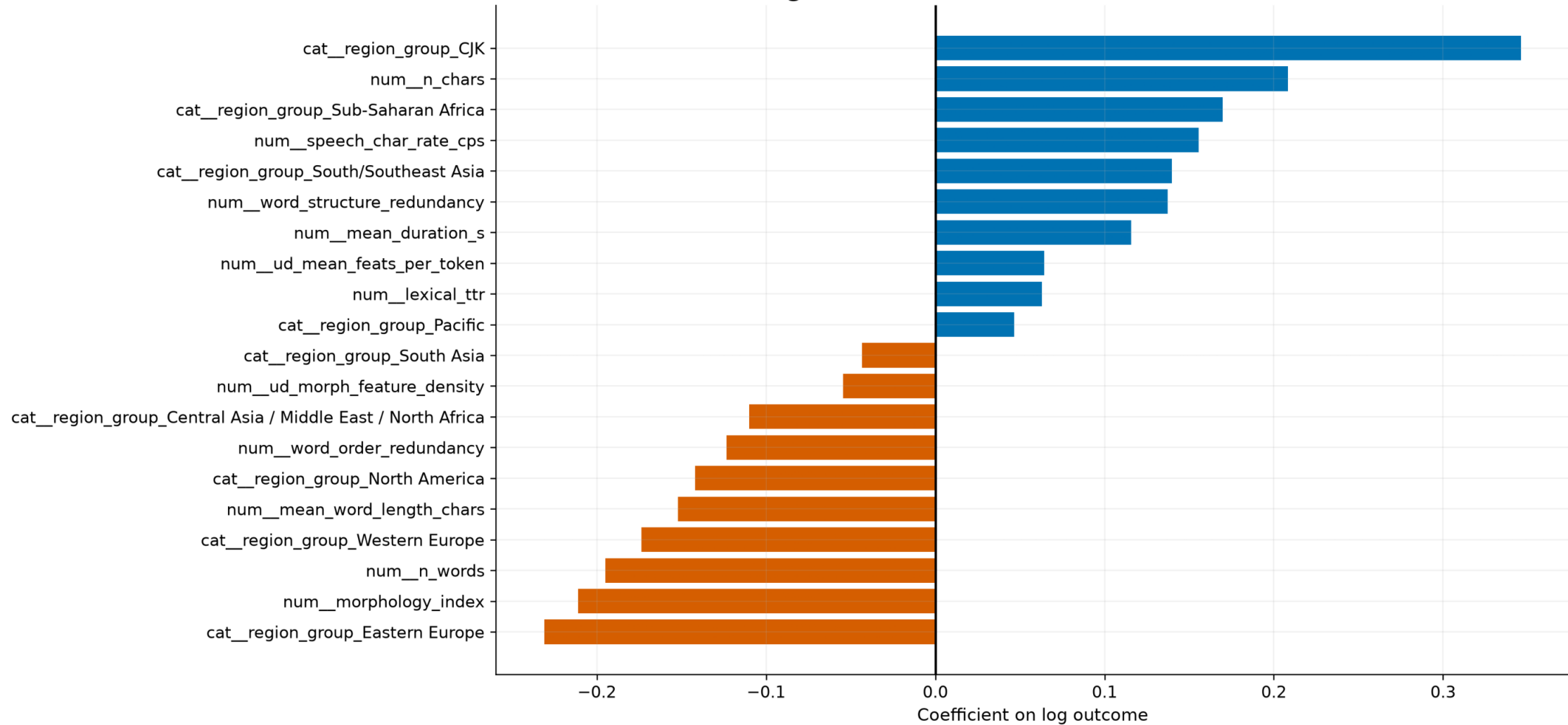
Caption — The 20 largest absolute leave-one-language-out residuals identify where the supplied predictor set misses language-specific locality variation.

## Largest LOCO-CV errors · time domain



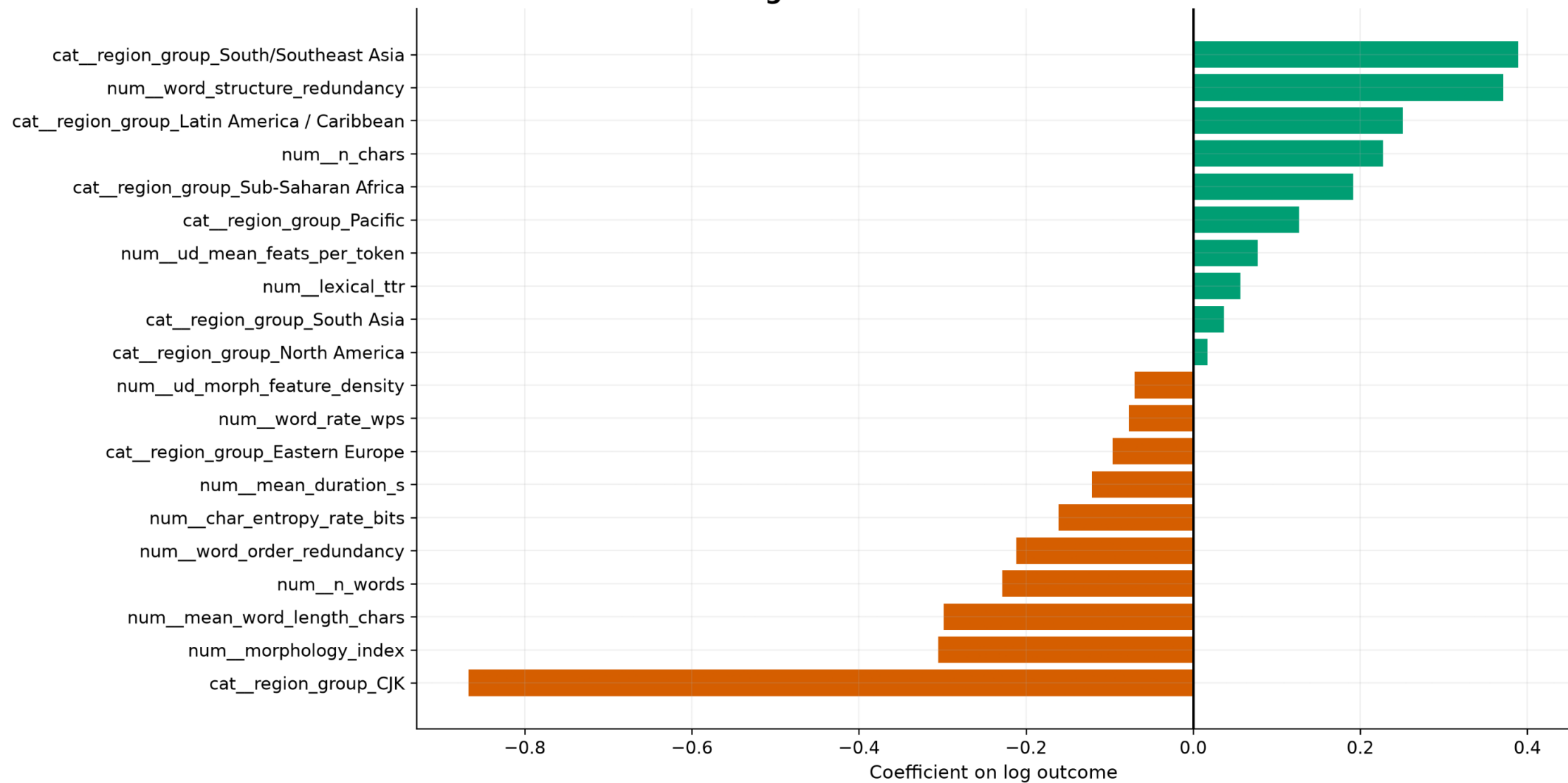
Caption — The 20 largest absolute leave-one-language-out residuals identify where the supplied predictor set misses language-specific locality variation.

## Largest fitted character-domain coefficients



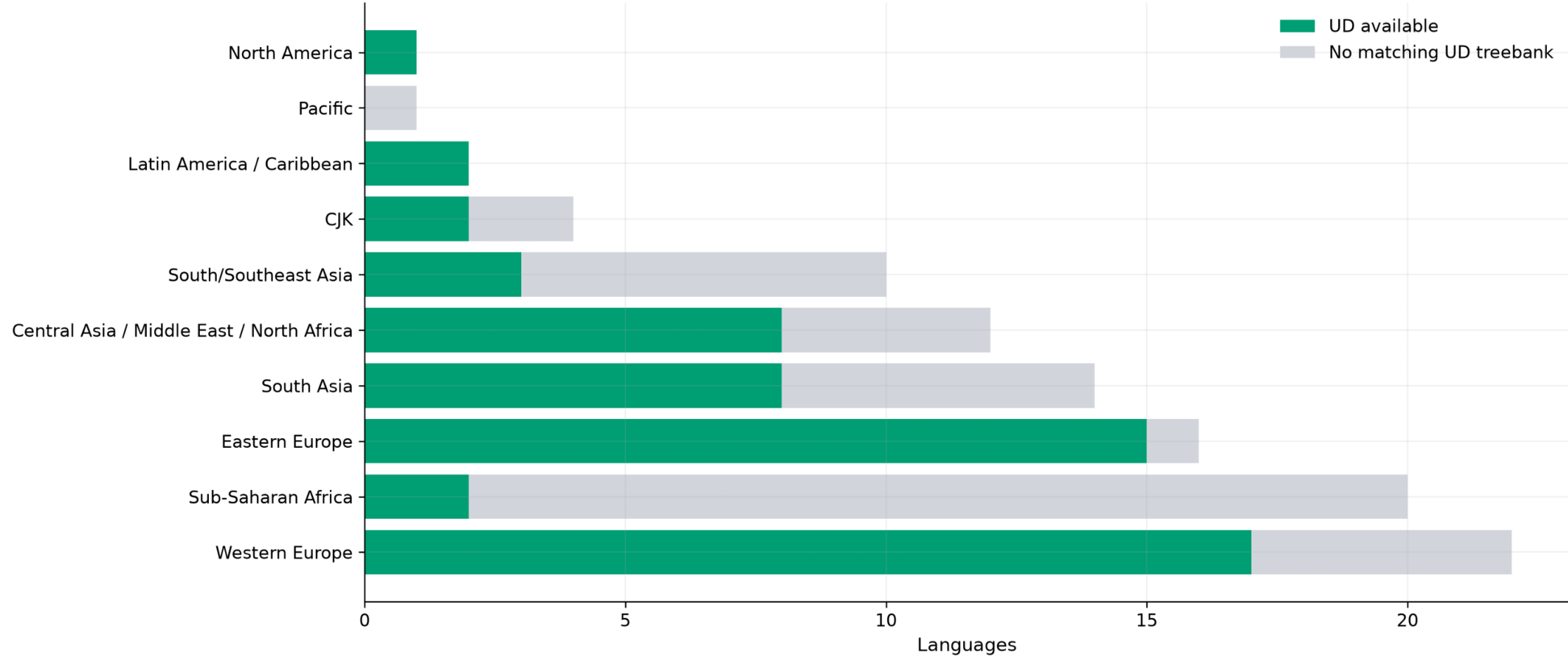
Caption — The largest positive and negative fitted coefficients are descriptive model parameters; the negative LOCO-CV  $R^2$  means they should not be read as validated causal or predictive effects.

## Largest fitted time-domain coefficients



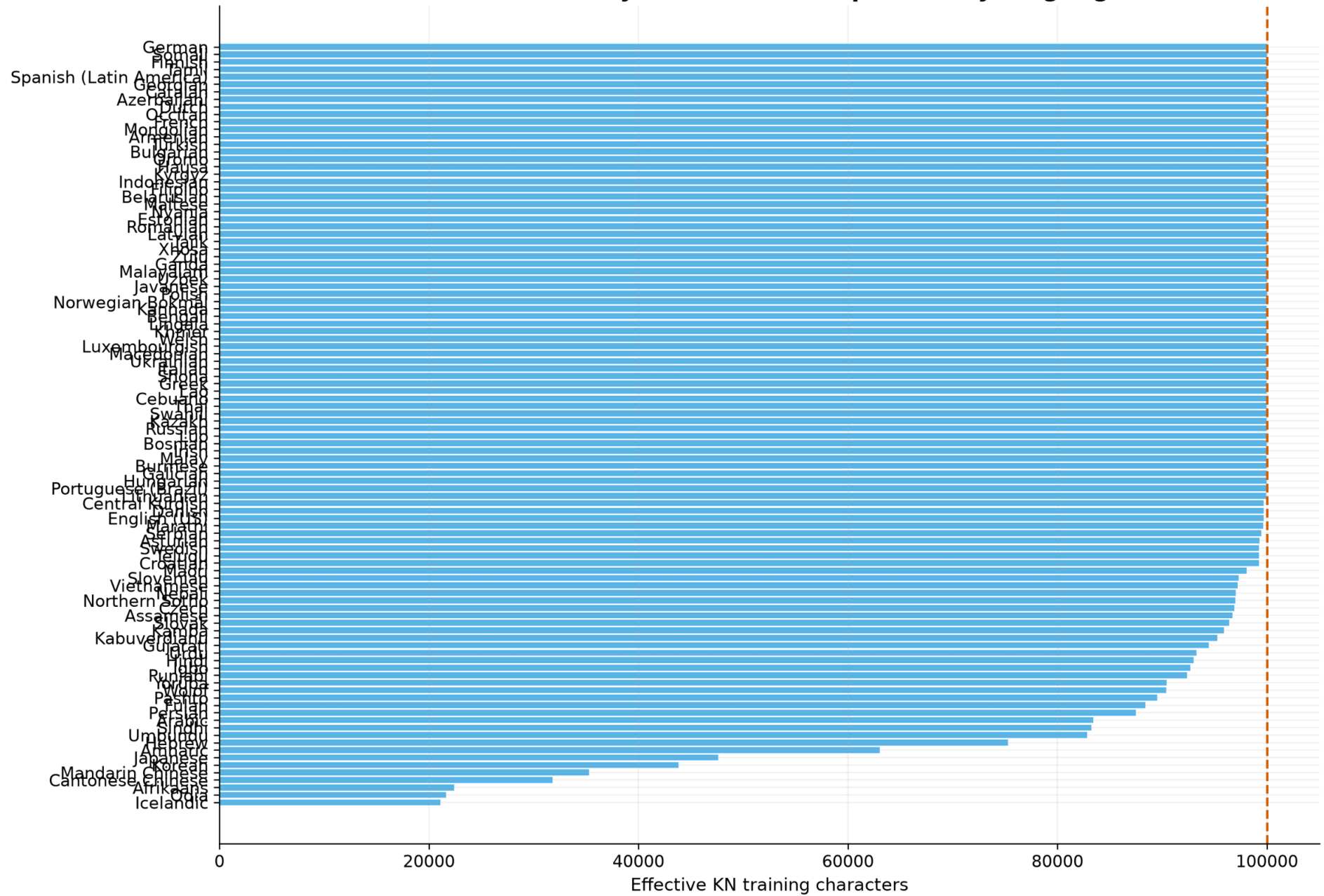
Caption — The largest positive and negative fitted coefficients are descriptive model parameters; the negative LOCO-CV  $R^2$  means they should not be read as validated causal or predictive effects.

## Structural-data coverage is uneven across the 102-language sample



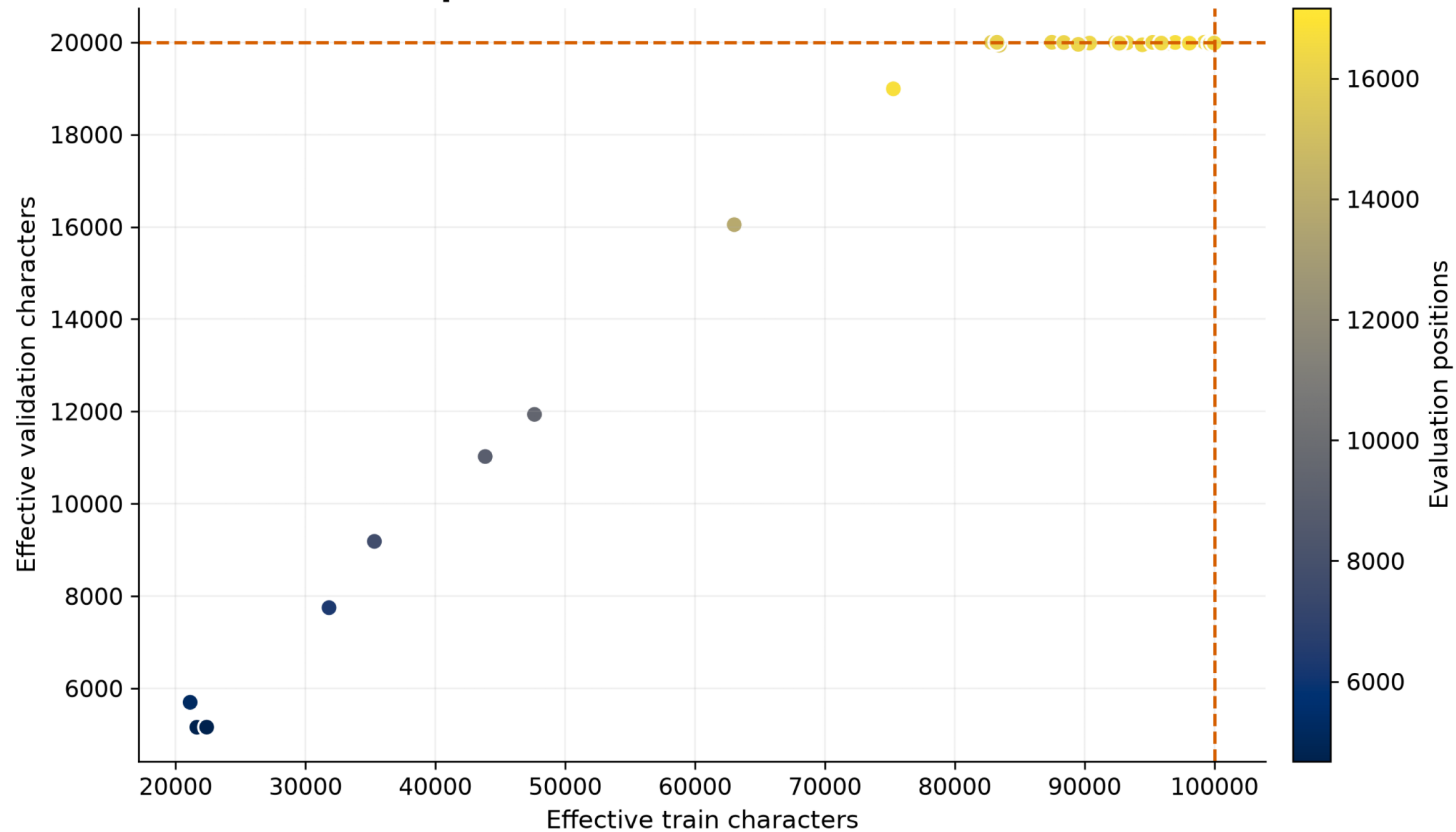
Caption — FLEURS locality is available for all 102 languages, but only 58 have matching UD annotations, so morphology analyses are necessarily a 58-language subset.

# Effective locality-estimation sample size by language



Caption — Effective KN training characters show where the adaptive estimator used less text than the original 100k upper cap, preventing low-resource languages from being rejected solely because they are smaller.

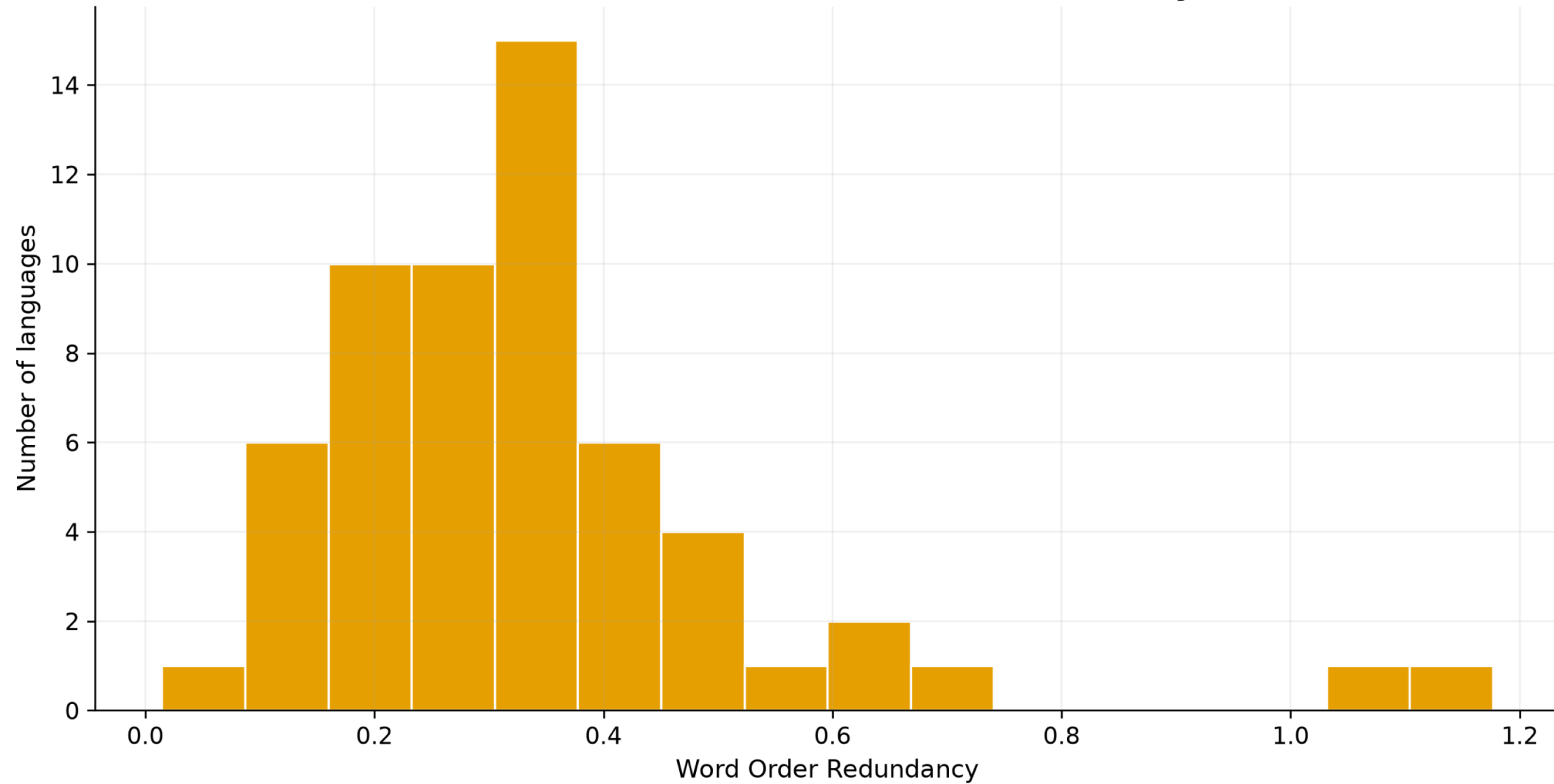
# Adaptive train/validation allocation



Caption — The dashed lines mark the originally requested 100k/20k caps; points below them show where actual language-level text availability forced a smaller but valid allocation.

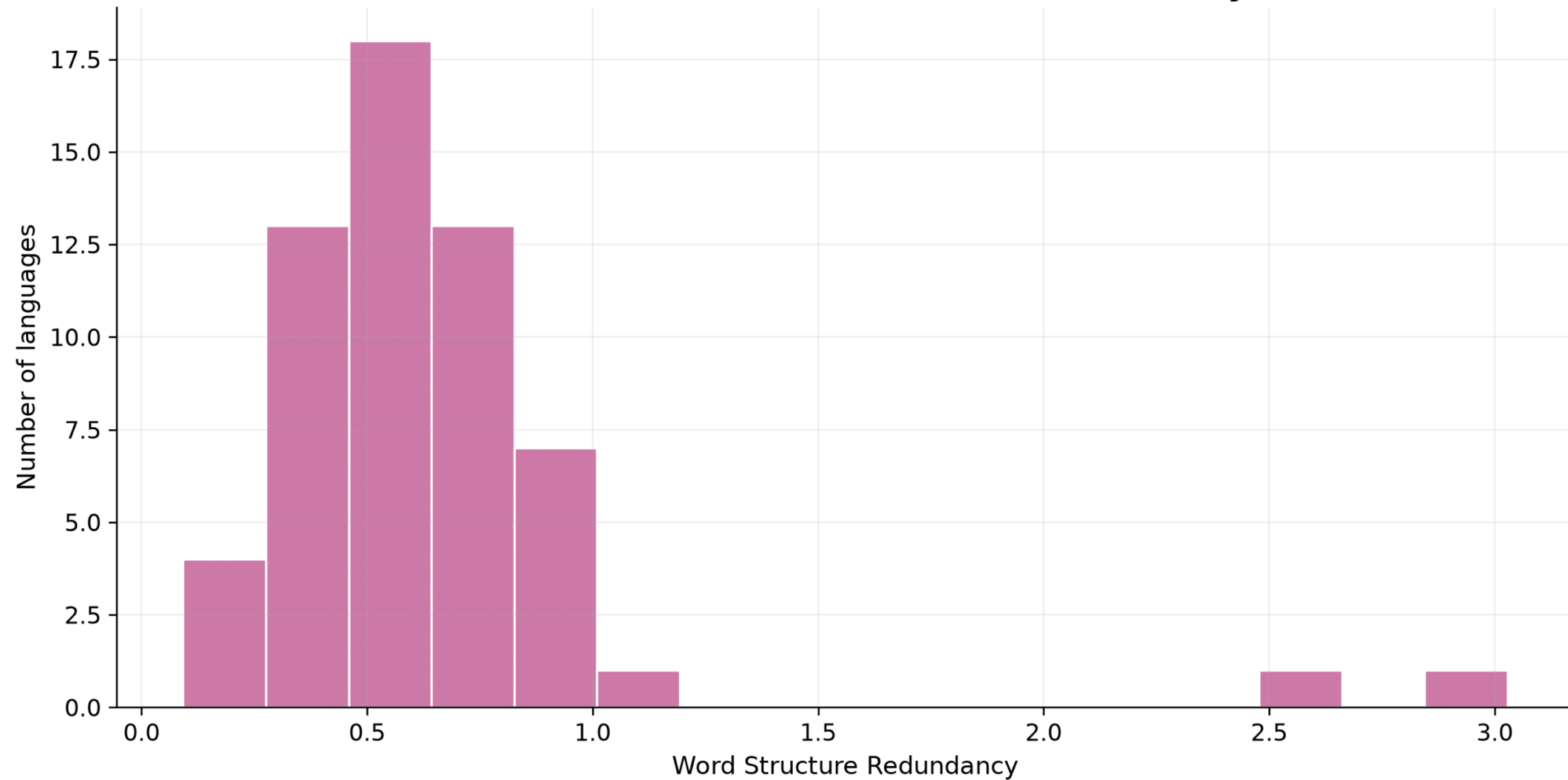


## Distribution of word-order redundancy



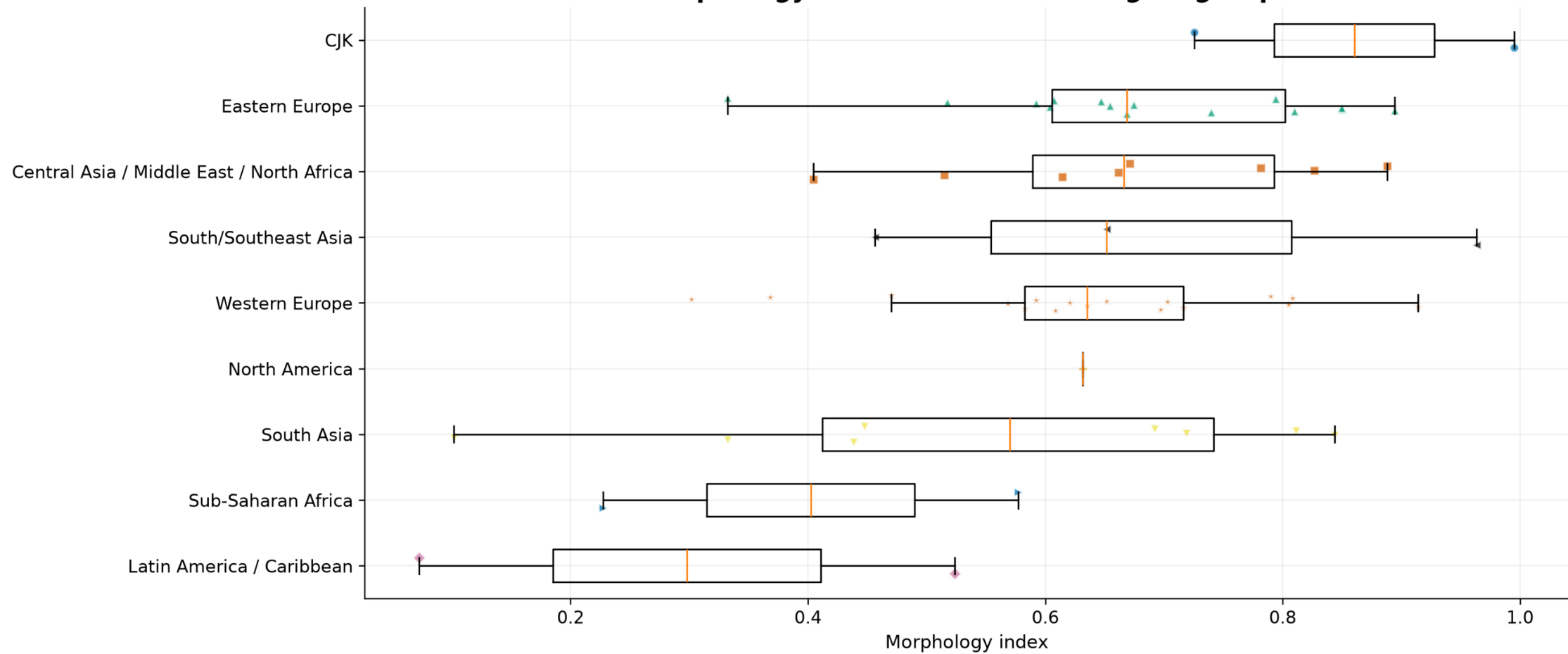
Caption — A direct distribution of one component of the analytic-synthetic trade-off, useful for seeing whether a few languages dominate the cross-language range.

## Distribution of word-structure redundancy



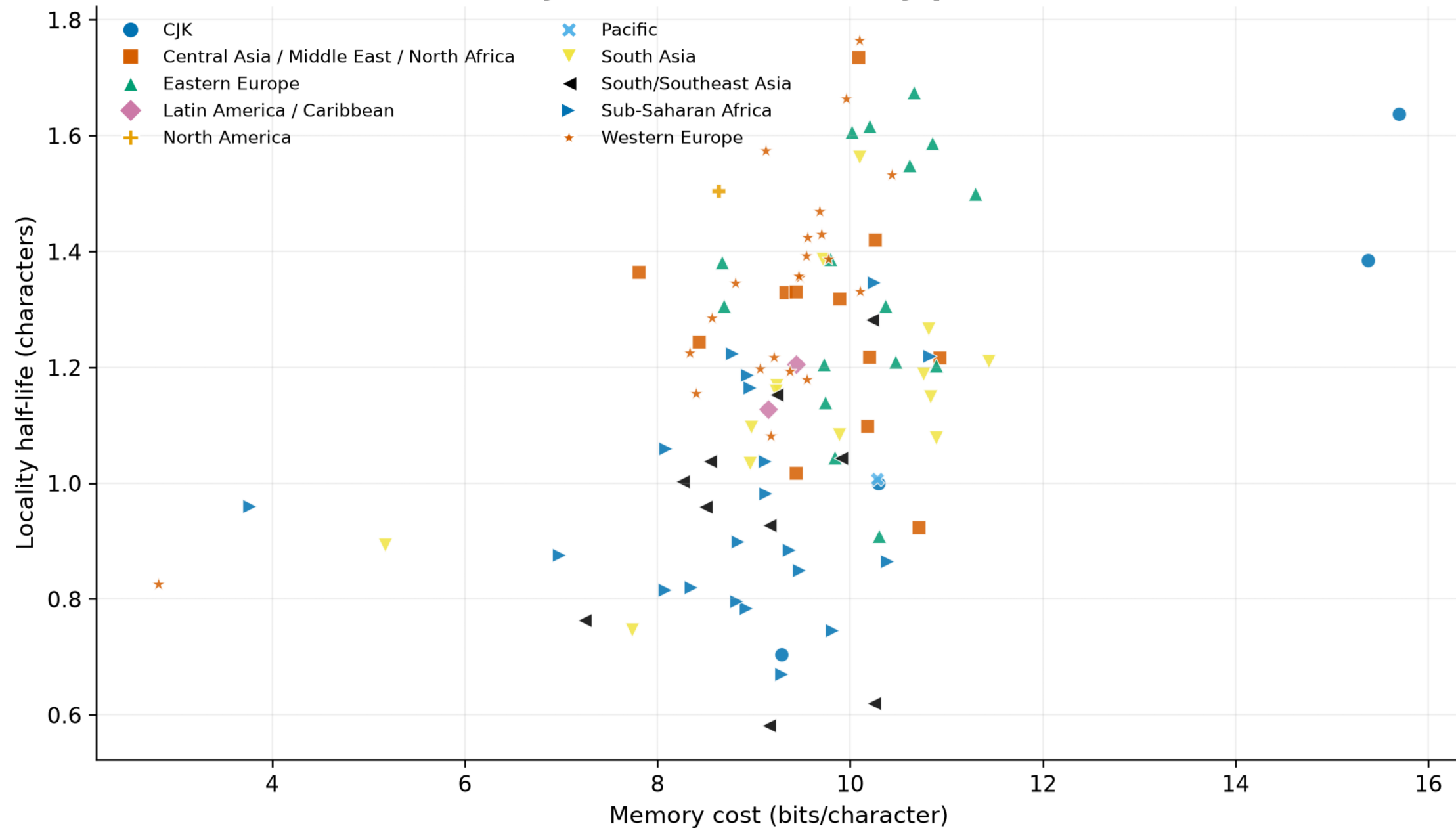
Caption — A direct distribution of one component of the analytic-synthetic trade-off, useful for seeing whether a few languages dominate the cross-language range.

## Morphology index varies across region groups



Caption — The morphology-index distribution differs across broad region groups, but this is an annotated 58-language subset with uneven UD coverage.

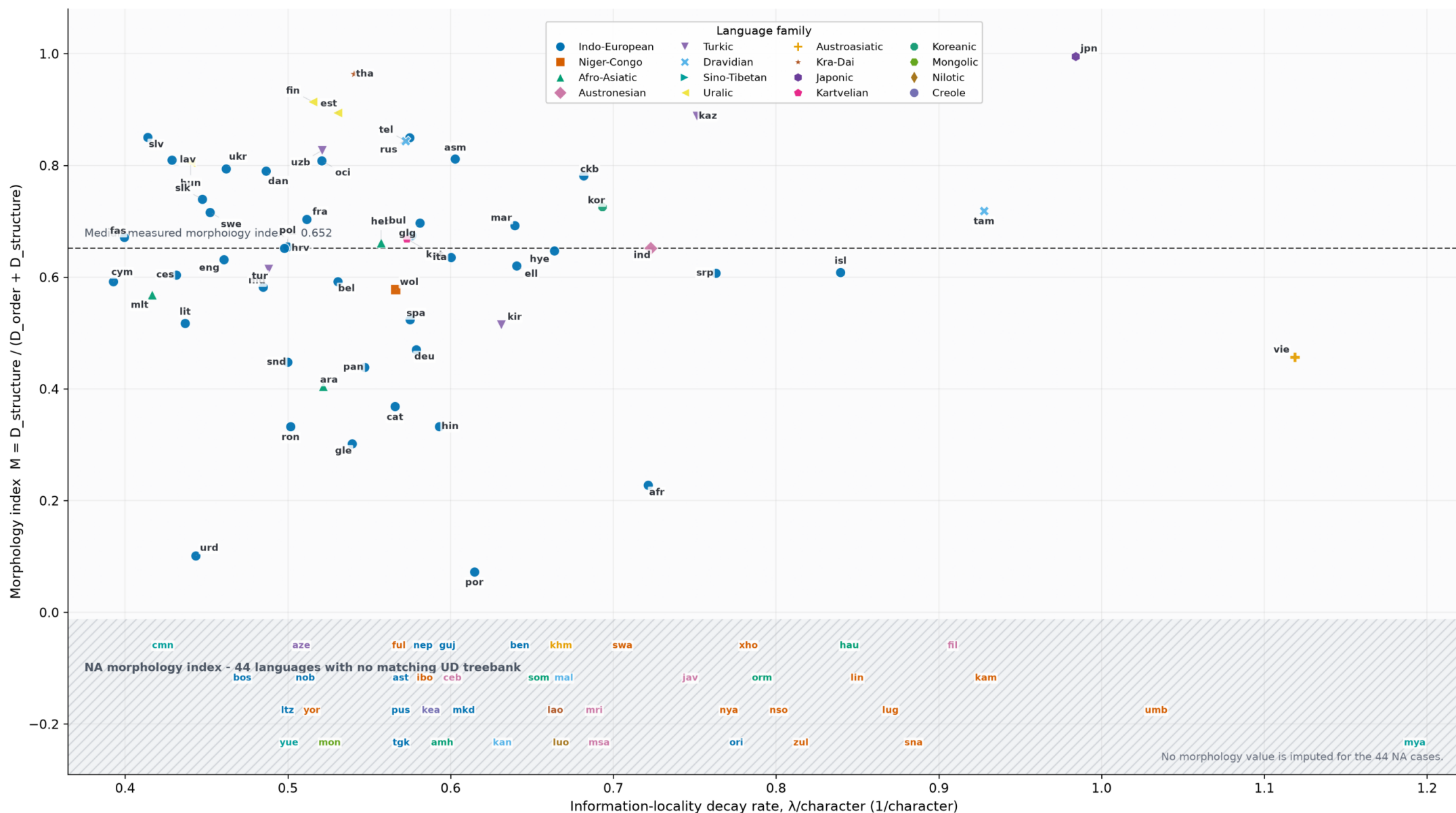
# Memory burden and locality persistence



Caption — A synthesis map combining cumulative memory cost with information half-life, giving a compact view of the study's efficiency-locality theme.

# Morphology-locality landscape across all 102 FLEURS languages: character domain

All 102 FLEURS languages | 58 measured morphology values + 44 UD-missing values | colour + marker = broad genealogical family



Caption — All 102 language rows are retained. The 58 languages with a matching UD treebank are plotted at their measured morphology index; the 44 languages without a matching UD treebank are shown in the hatched NA strip at their observed character-domain decay coordinate. No morphology value is imputed.

# Morphology-locality landscape across all 102 FLEURS languages: speech-time domain

All 102 FLEURS languages | 58 measured morphology values + 44 UD-missing values | colour + marker = broad genealogical family



Caption — All 102 language rows are retained. The 58 languages with a matching UD treebank are plotted at their measured morphology index; the 44 languages without a matching UD treebank are shown in the hatched NA strip at their observed speech-time decay coordinate. No morphology value is imputed.